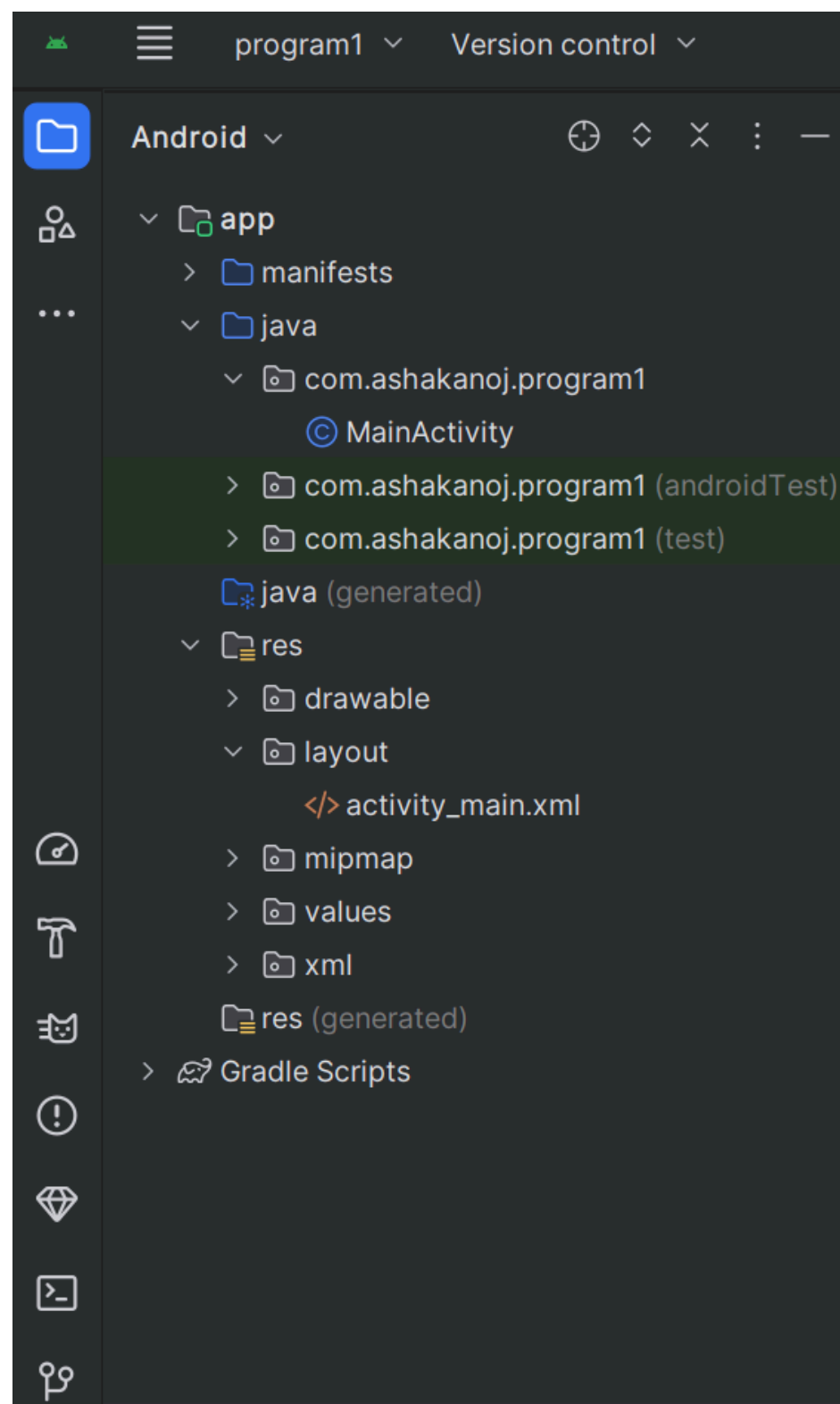


List of Lab Programs

PART- A

1. Creating “Hello world” Application.

STRUCTURE OF A PROGRAM:



activity_main.xml file:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:background="@color/green"
tools:context=".MainActivity">

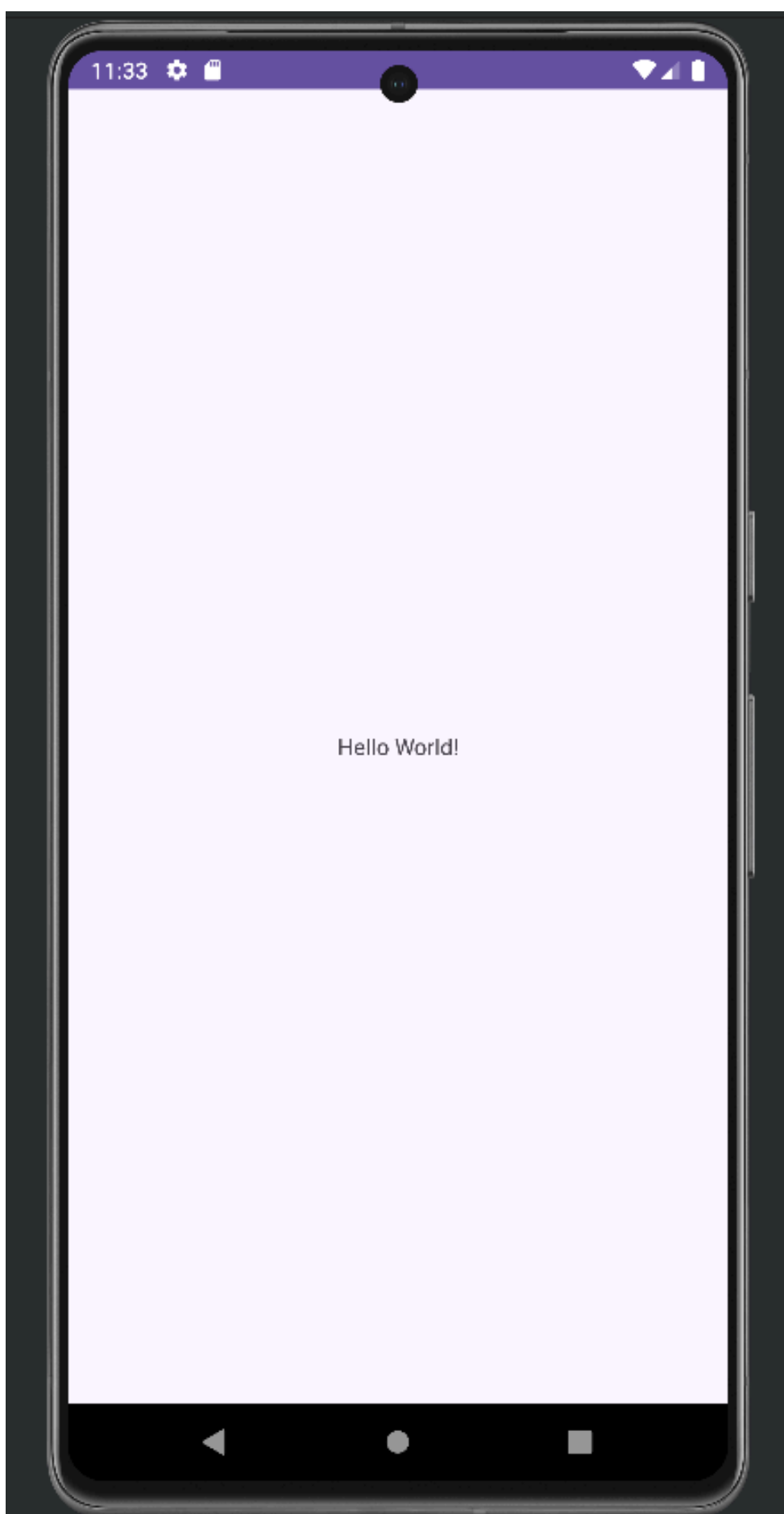
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        android:textSize="50dp"
        android:textStyle="italic"
        android:textColor="@color/white"
        android:padding="50dp"
        app:layout_constraintBottom_toBottomOf="parent"
```

```
        app:layout_constraintEnd_toEndOf="parent"  
        app:layout_constraintStart_toStartOf="parent"  
        app:layout_constraintTop_toTopOf="parent" />  
</androidx.constraintlayout.widget.ConstraintLayout>
```

Mainactivity.java file:

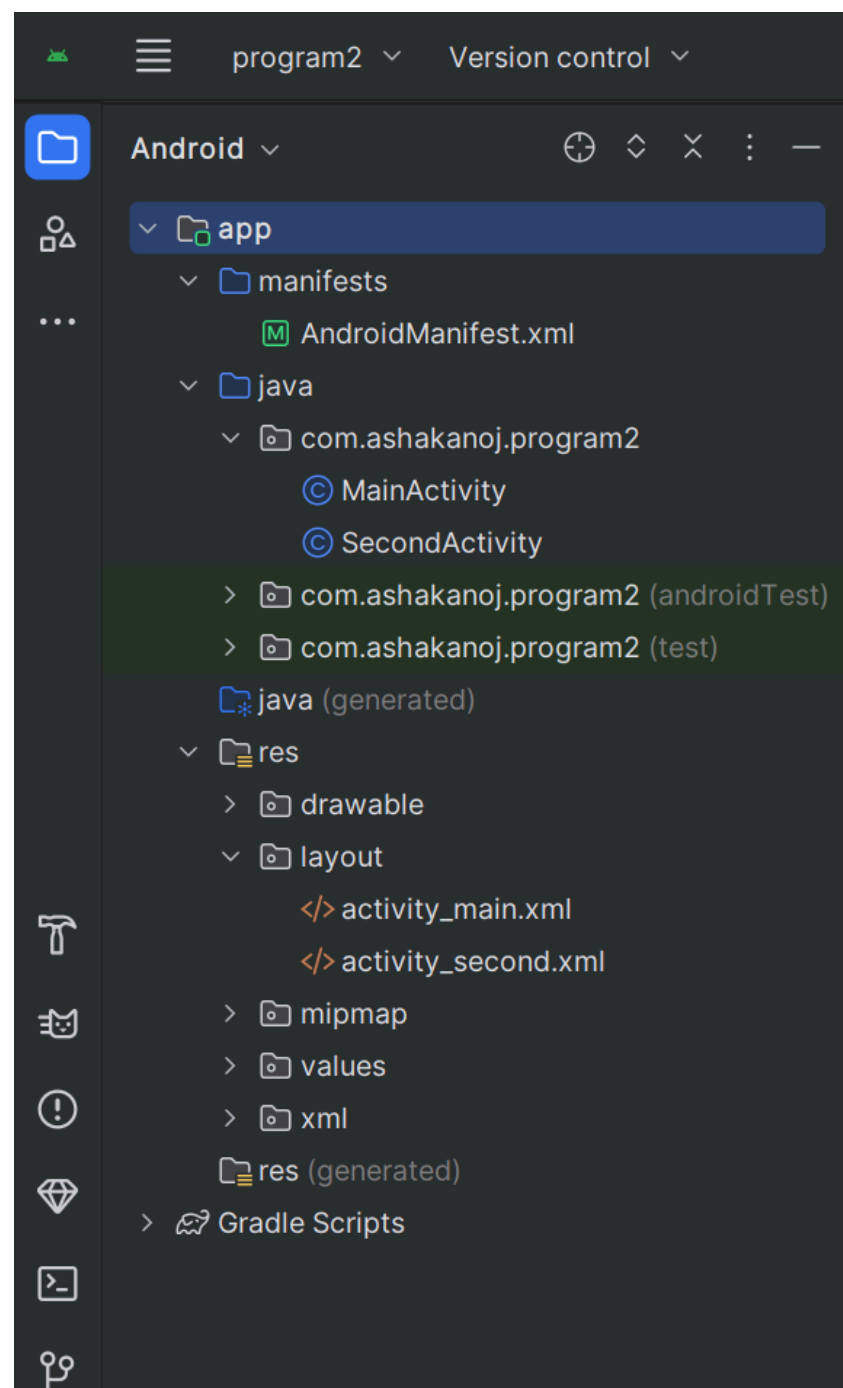
```
package com.ashakanoj.program1;  
  
import androidx.appcompat.app.AppCompatActivity;  
import android.os.Bundle;  
  
public class MainActivity extends AppCompatActivity {  
  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        setContentView(R.layout.activity_main);  
    }  
}
```

Output:



2. Creating an application that displays message based on the screen orientation.

STRUCTURE OF A PROGRAM:



AndroidManifest.xml file:

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Screenorientation"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        <activity android:name=".SecondActivity"
            android:screenOrientation="landscape">
        </activity>
    </application>
```

</manifest>

activity_main.xml file:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="8dp"
        android:layout_marginTop="112dp"
        android:onClick="onClick"
        android:text="Launch next activity"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.612"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/editText1"
        app:layout_constraintVertical_bias="0.613" />

    <TextView
        android:id="@+id/editText1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerHorizontal="true"
        android:layout_marginEnd="8dp"
        android:layout_marginStart="8dp"
        android:layout_marginTop="124dp"
        android:ems="10"
        android:textSize="22dp"
        android:text="This activity is portrait orientation"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.502"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

activity_second.xml file:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".SecondActivity">
```



```

<TextView
    android:id="@+id/textView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginEnd="8dp"
    android:layout_marginStart="8dp"
    android:layout_marginTop="180dp"
    android:text="This is landscape orientation"
    android:textSize="22dp"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.502"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

MainActivity.java file:

```

package com.ashakanoj.program2;

import androidx.appcompat.app.AppCompatActivity;
import android.view.View;
import android.widget.Button;
import android.content.Intent;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {
    Button button1;

    @Override

    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        button1 = (Button) findViewById(R.id.button1);
    }

    public void onClick(View v) {
        Intent intent = new Intent(MainActivity.this, SecondActivity.class);
        startActivity(intent);
    }
}

```

SecondActivity.java file:

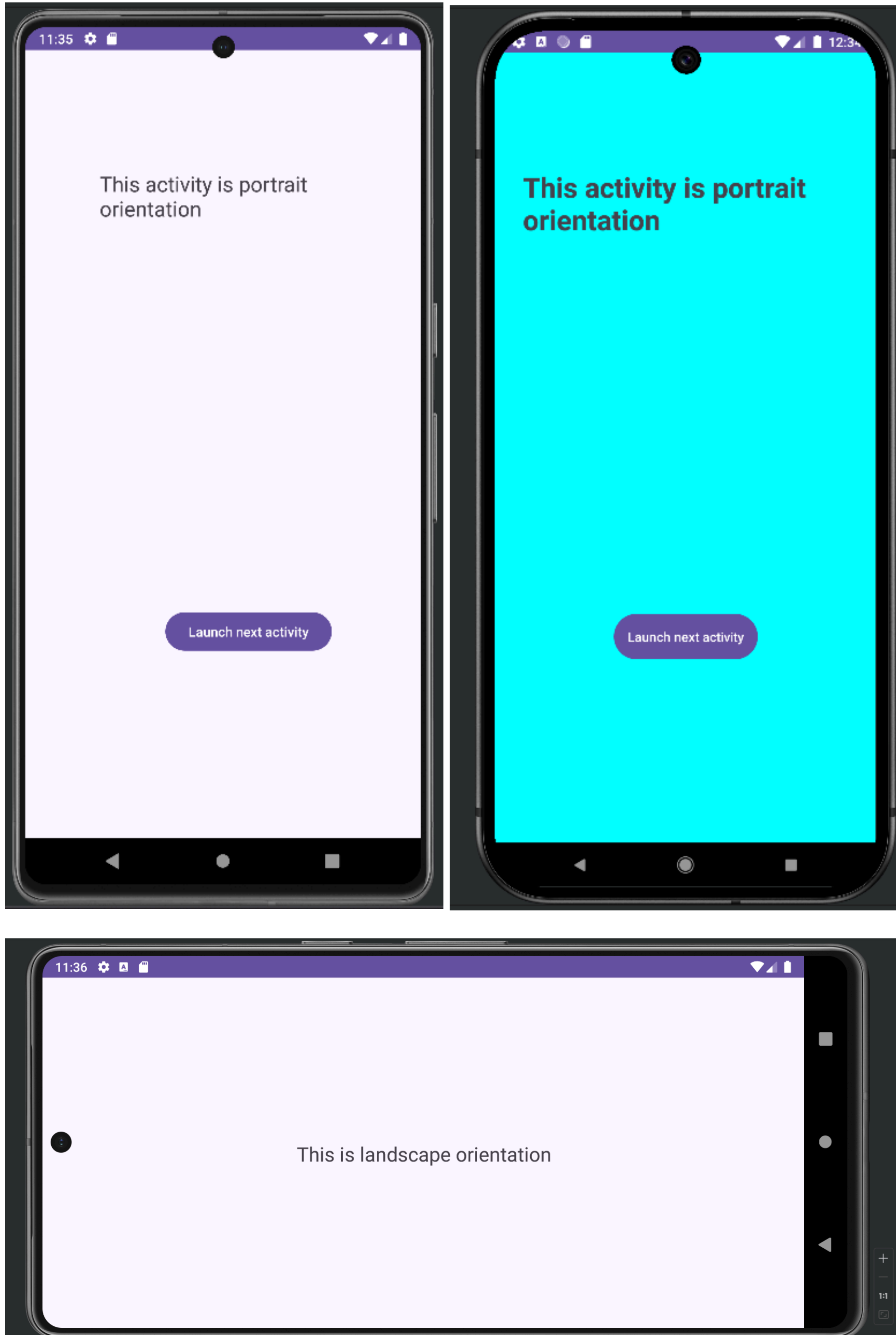
```

package com.ashakanoj.program2;
import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);
    }
}

```

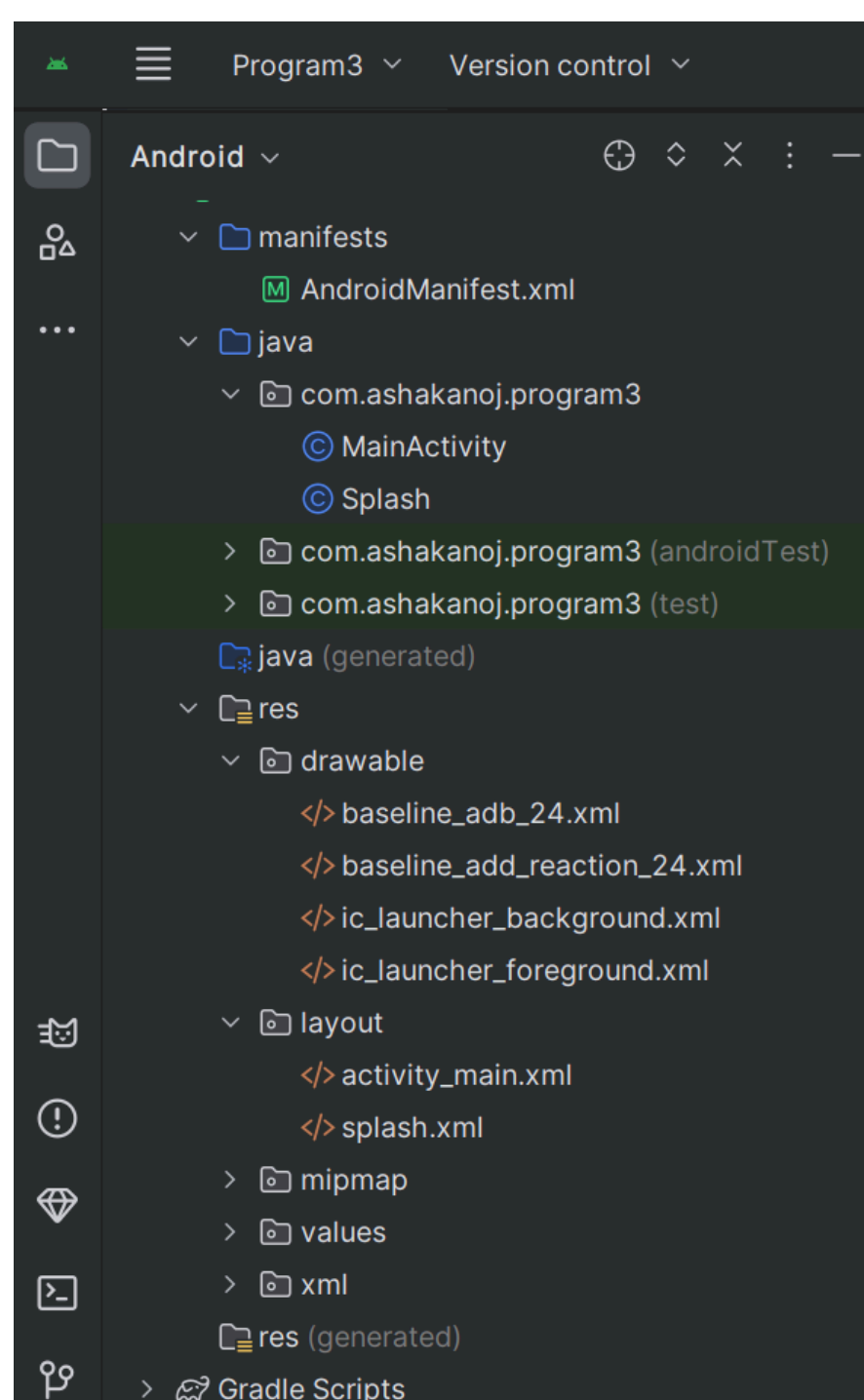
Output:



3. Create an application that displays custom designed Opening Screen.

<https://abhiandroid.com/programming/splashscreen#gsc.tab=0>

STRUCTURE OF A PROGRAM:



AndroidManifest.xml file:

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:icon="@drawable/baseline_add_reaction_24"
        android:label="@string/app_name"
        android:supportsRtl="true"
```



```

        android:theme="@style/Theme.AppCompat.DayNight">
        <activity android:name="com.ashakanoj.program3.Splash"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        <activity android:name="com.ashakanoj.program3.MainActivity"/>
    </application>

</manifest>

```

activity_main.xml file:

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/activity_main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welocme to Opening Screen!!"
        android:textSize="20sp"
        android:layout_centerInParent="true"/>
</RelativeLayout>

```

splash.xml file:

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:background="@color/design_default_color_primary_variant">

    <ImageView
        android:id="@+id/logo_id"
        android:layout_width="250dp"
        android:layout_height="250dp"
        android:layout_centerInParent="true"
        android:src="@drawable/baseline_add_reaction_24"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@+id/logo_id"
        android:layout_centerHorizontal="true"
        android:text="This is an Splash Screen"
        android:textSize="40dp"
        android:textColor="@color/black"/>

```

```
</RelativeLayout>
```

MainActivity.java file:

```
package com.ashakanoj.program3;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Splash.java file:

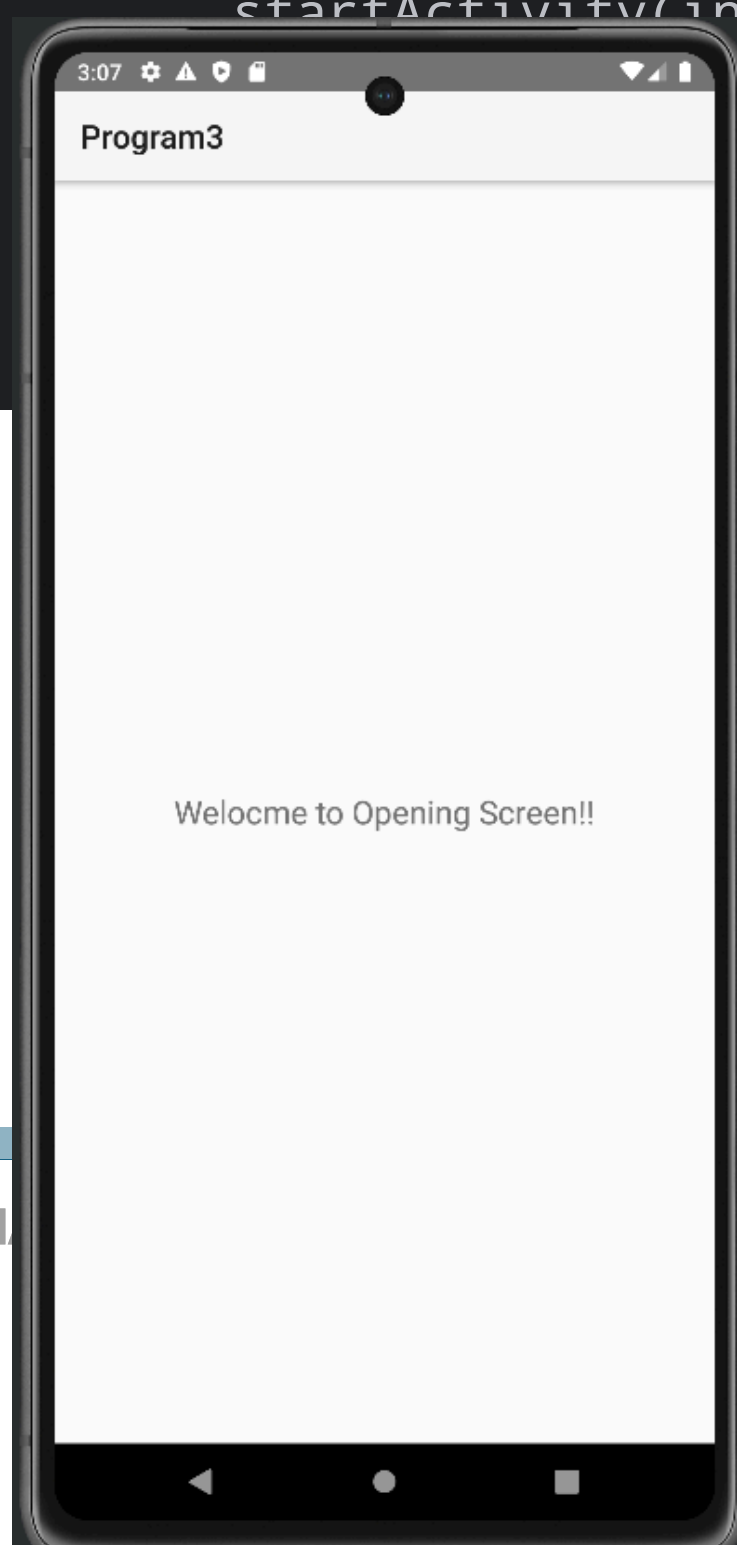
```
package com.ashakanoj.program3;

import android.app.Activity;
import android.content.Intent;
import android.os.Bundle;
import android.os.Handler;

import java.nio.Buffer;
//import java.util.logging.Handler;

public class Splash extends Activity {
    Handler handler;
    @Override
    protected void onCreate(Bundle savedInstanceState){
        super.onCreate(savedInstanceState);
        setContentView(R.layout.splash);
        handler = new Handler();
        handler.postDelayed(new Runnable() {
            @Override
            public void run() {
                Intent intent = new Intent(Splash.this,
MainActivity.class);
                startActivity(intent);
                finish();
            }
        },3000);
    }
}
```

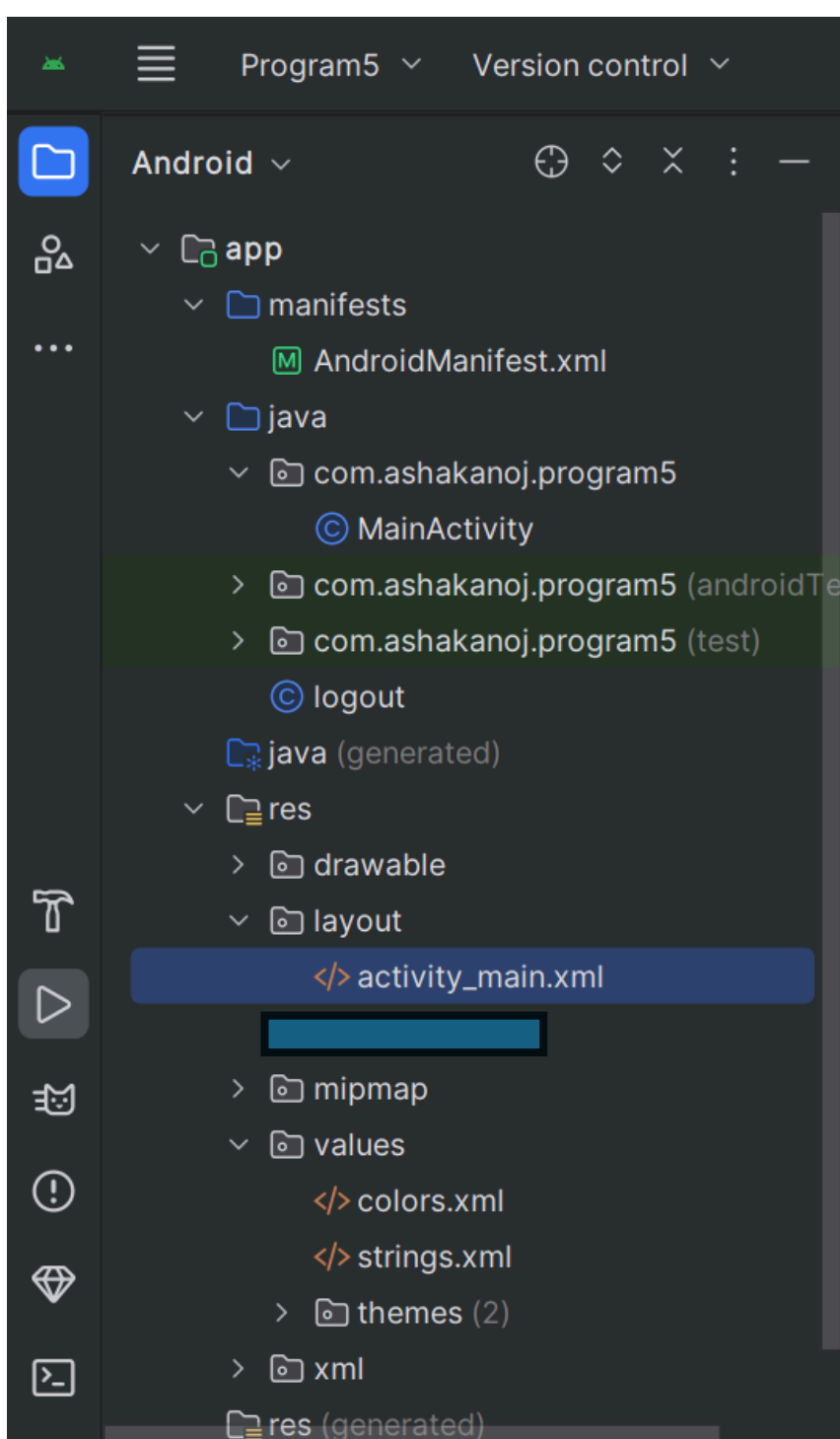
Output:





4. Create an application to develop Login window using UI controls.
<https://abhiandroid.com/ui/absolute-layout#gsc.tab=0>

STRUCTURE OF A PROGRAM:



Activity_main.xml File:

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/img_4"
    tools:context=".MainActivity">

    <TextView android:text = "Login Page"
        android:layout_width="wrap_content"
        android:layout_height = "wrap_content"
        android:id = "@+id/textview"
        android:textSize = "40dp"
        android:textColor="@color/black"
        android:layout_alignParentTop = "true"
        android:layout_centerHorizontal = "true" />

    <TextView
        android:layout_width = "wrap_content"
        android:layout_height = "wrap_content"
        android:text = "Surana College"
        android:id = "@+id/textView"
        android:layout_below = "@+id/textview"
        android:layout_centerHorizontal = "true"
        android:textColor = "#ff7aff24"
        android:textSize = "35dp" />

    <EditText
        android:layout_width = "wrap_content"
        android:layout_height = "wrap_content"
        android:id = "@+id/editText"
        android:hint = "Enter Name"
        android:focusable = "true"
        android:textColorHighlight = "#ff7eff15"
        android:textColorHint = "#FFFFFFFF"
        android:layout_marginTop = "46dp"
        android:layout_below = "@+id/imageView"
        android:layout_alignParentLeft = "true"
        android:layout_alignParentStart = "true"
        android:layout_alignParentRight = "true"
        android:layout_alignParentEnd = "true" />

    <ImageView
        android:layout_width="wrap_content"
        android:layout_height="200dp"
        android:id="@+id/imageView"
        android:src="@drawable/img"
        android:layout_below="@+id/textView"
        android:layout_centerHorizontal="true" />

    <EditText
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:inputType="textPassword"

```



```

        android:ems="10"
        android:id="@+id/editText2"
        android:layout_below="@+id/editText"
        android:layout_alignParentLeft="true"
        android:layout_alignParentStart="true"
        android:layout_alignRight="@+id/editText"
        android:layout_alignEnd="@+id/editText"
        android:textColorHint="#FFFFFFFF"
        android:hint="Password" />

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Attempts Left:"
    android:id="@+id/textView2"
    android:layout_below="@+id/editText2"
    android:layout_alignParentLeft="true"
    android:layout_alignParentStart="true"
    android:textColor="@color/black"
    android:textSize="25dp" />

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="New Text"
    android:id="@+id/textView3"
    android:layout_alignTop="@+id/textView2"
    android:layout_alignParentRight="true"
    android:layout_alignParentEnd="true"
    android:layout_alignBottom="@+id/textView2"
    android:layout_toEndOf="@+id/textview"
    android:textSize="25dp"
    android:textColor="@color/black"
    android:layout_toRightOf="@+id/textview" />

<Button
    android:layout_width="wrap_content"
    android:layout_height="60dp"
    android:text="login"
    android:id="@+id/button"
    android:layout_alignParentBottom="true"
    android:layout_toLeftOf="@+id/textview"
    android:layout_marginTop="100dp"
    android:layout_toStartOf="@+id/textview" />

<Button
    android:layout_width="wrap_content"
    android:layout_height="60dp"
    android:text="Cancel"
    android:id="@+id/button2"
    android:layout_alignParentBottom="true"
    android:layout_toRightOf="@+id/textview"
    android:layout_toEndOf="@+id/textview" />

</RelativeLayout>

```


MainActivity.java File:

```

package com.ashakanoj.program5;

import androidx.appcompat.app.AppCompatActivity;

import android.content.Intent;
import android.os.Bundle;
import android.graphics.Color;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    Button b1,b2;
    EditText ed1,ed2;

    TextView tx1;
    int counter = 3;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        b1 = (Button)findViewById(R.id.button);
        ed1 = (EditText)findViewById(R.id.editText);
        ed2 = (EditText)findViewById(R.id.editText2);

        b2 = (Button)findViewById(R.id.button2);
        tx1 = (TextView)findViewById(R.id.textView3);
        tx1.setVisibility(View.GONE);

        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                if(ed1.getText().toString().equals("admin") &&
                    ed2.getText().toString().equals("admin")) {
                    Toast.makeText(getApplicationContext(),
                        "Redirecting...Login
Successful !!!",Toast.LENGTH_SHORT).show();

                }else{
                    Toast.makeText(getApplicationContext(), "Wrong
Credentials",Toast.LENGTH_SHORT).show();

                    tx1.setVisibility(View.VISIBLE);
                    tx1.setBackgroundColor(Color.RED);
                    counter--;
                    tx1.setText(Integer.toString(counter));

                    if (counter == 0) {
                        b1.setEnabled(false);

```

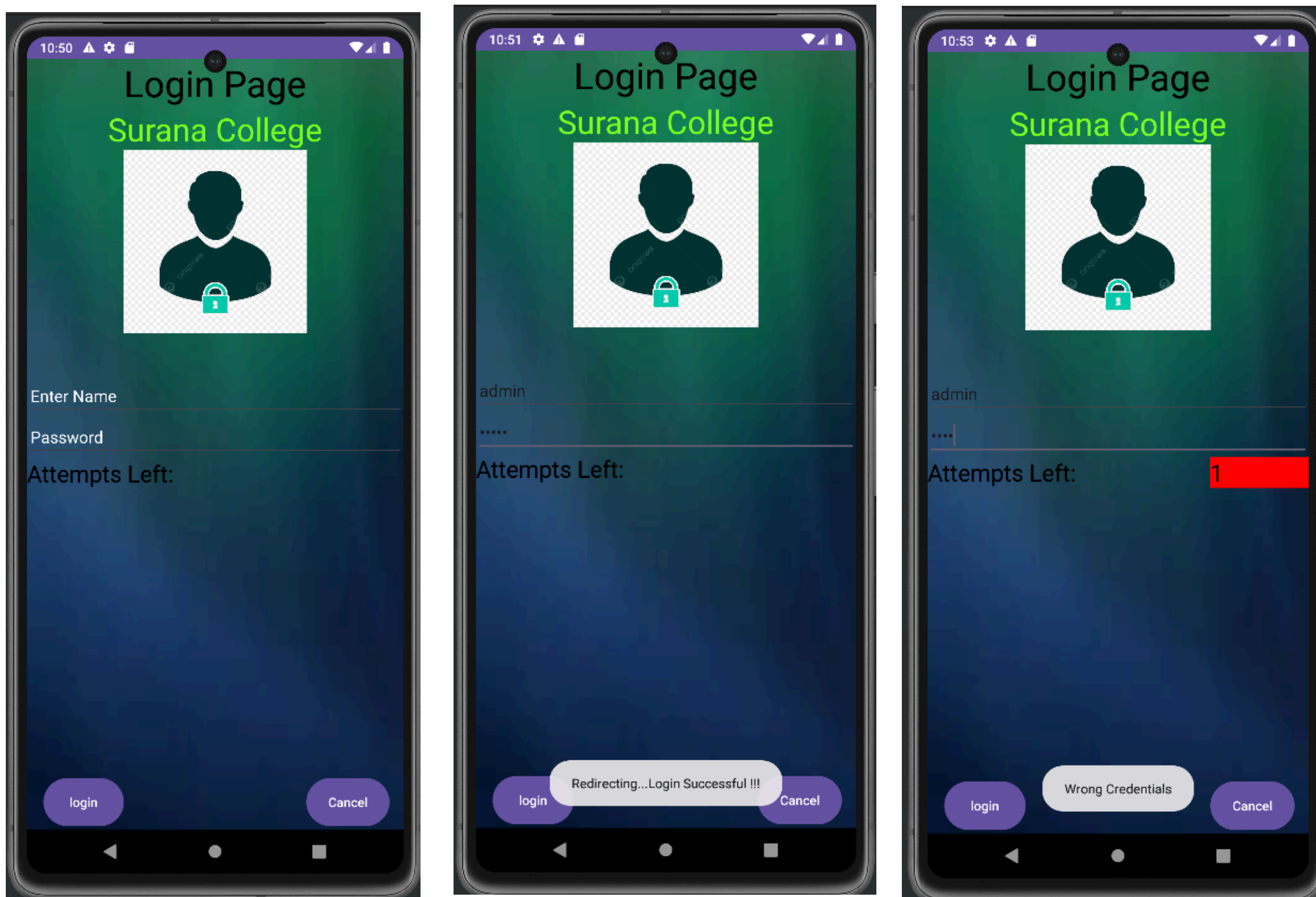
```

    }
    }
    });

    b2.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {

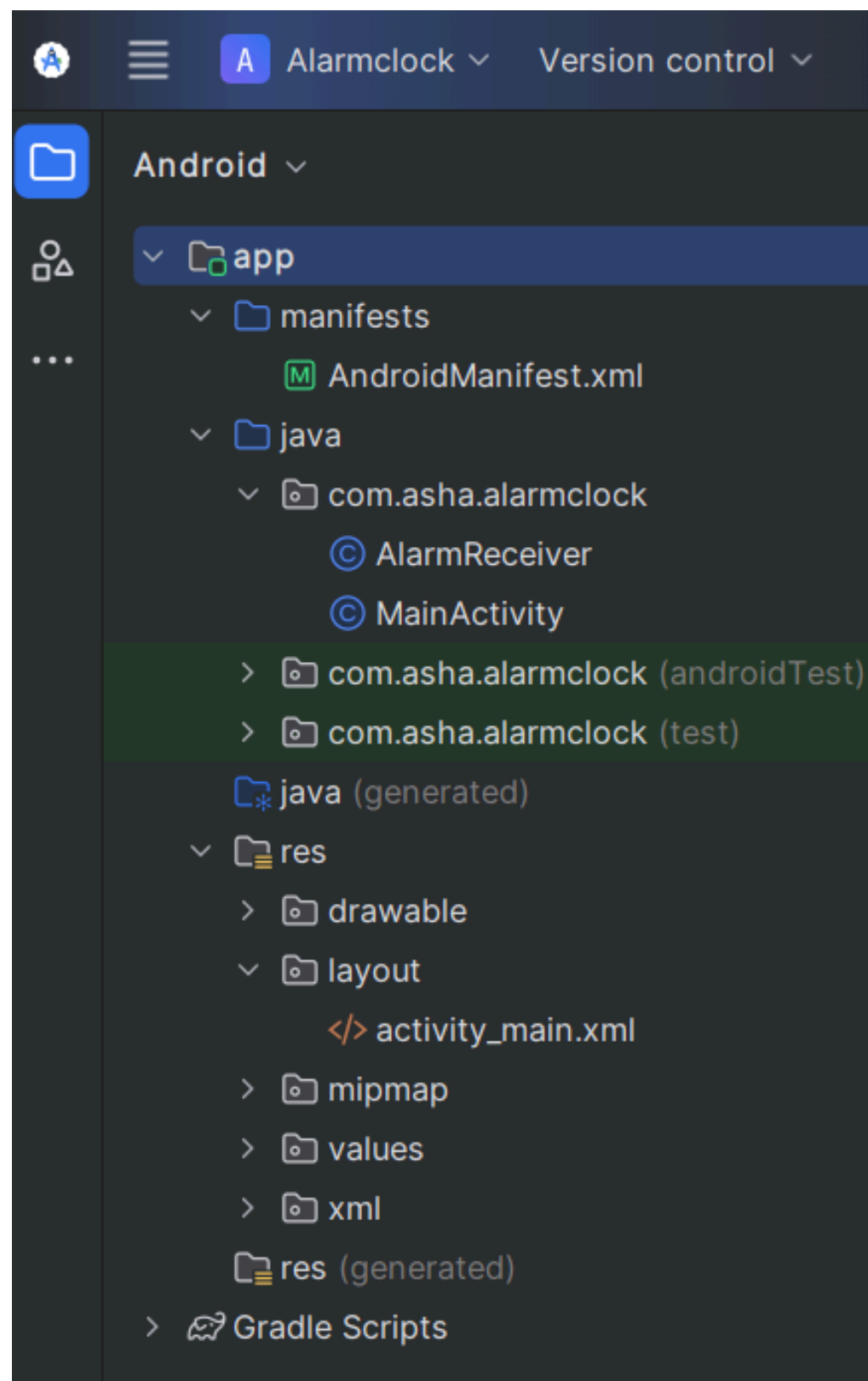
        }
    });
}
}

```

Output:

5. Create Alarm clock in Android Application.

STRUCTURE OF A PROGRAM:



AndroidManifest.xml File:

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <uses-permission android:name="android.permission.SET_ALARM"/>
    <uses-permission android:name="android.permission.WAKE_LOCK"/>

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/AppTheme"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>
```



```

        </intent-filter>
    </activity>
    <receiver android:name=".AlarmReceiver"
        android:exported="false"/>
</application>
</manifest>

```

activity_main.xml File:

```

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <DatePicker
        android:id="@+id/datePicker"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <TimePicker
        android:id="@+id/timePicker"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:timePickerMode="spinner"/>

    <Button
        android:id="@+id/setAlarmButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Set Alarm" />

    <Button
        android:id="@+id/stopAlarmButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Stop Alarm"
        android:visibility="gone"/> <!-- Initially hidden -->
</LinearLayout>

```

MainActivity.java File:

```

package com.asha.alarmclock;

import android.app.AlarmManager;
import android.app.PendingIntent;
import android.content.Context;
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.DatePicker;
import android.widget.TimePicker;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import java.util.Calendar;

```



```

public class MainActivity extends AppCompatActivity {

    private DatePicker datePicker;
    private TimePicker timePicker;
    private Button setAlarmButton, stopAlarmButton;
    private AlarmManager alarmManager;
    private PendingIntent pendingIntent;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        datePicker = findViewById(R.id.datePicker);
        timePicker = findViewById(R.id.timePicker);
        setAlarmButton = findViewById(R.id.setAlarmButton);
        stopAlarmButton = findViewById(R.id.stopAlarmButton);

        alarmManager = (AlarmManager) getSystemService(Context.ALARM_SERVICE);

        setAlarmButton.setOnClickListener(v -> setAlarm());
        stopAlarmButton.setOnClickListener(v -> stopAlarm());
    }

    private void setAlarm() {
        int year = datePicker.getYear();
        int month = datePicker.getMonth();
        int day = datePicker.getDayOfMonth();
        int hour = timePicker.getHour();
        int minute = timePicker.getMinute();

        Calendar calendar = Calendar.getInstance();
        calendar.set(year, month, day, hour, minute, 0);

        Intent intent = new Intent(this, AlarmReceiver.class);
        pendingIntent = PendingIntent.getBroadcast(this, 0, intent,
PendingIntent.FLAG_UPDATE_CURRENT);

        alarmManager.setExact(AlarmManager.RTC_WAKEUP, calendar.getTimeInMillis(),
pendingIntent);

        stopAlarmButton.setVisibility(Button.VISIBLE); // Show stop button

        Toast.makeText(this, "Alarm Set for " + day + "/" + (month + 1) + "/" + year + " "
+ hour + ":" + minute, Toast.LENGTH_SHORT).show();
    }

    private void stopAlarm() {
        AlarmReceiver.stopRingtone(); // Stop the ringing

        if (pendingIntent != null) {
            alarmManager.cancel(pendingIntent);
            pendingIntent.cancel();
        }

        stopAlarmButton.setVisibility(Button.GONE); // Hide stop button
        Toast.makeText(this, "Alarm Stopped", Toast.LENGTH_SHORT).show();
    }
}

```


AlarmReceiver.java File:

```
package com.asha.alarmclock;

import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.media.Ringtone;
import android.media.RingtoneManager;
import android.net.Uri;
import android.widget.Toast;

public class AlarmReceiver extends BroadcastReceiver {

    private static Ringtone ringtone;

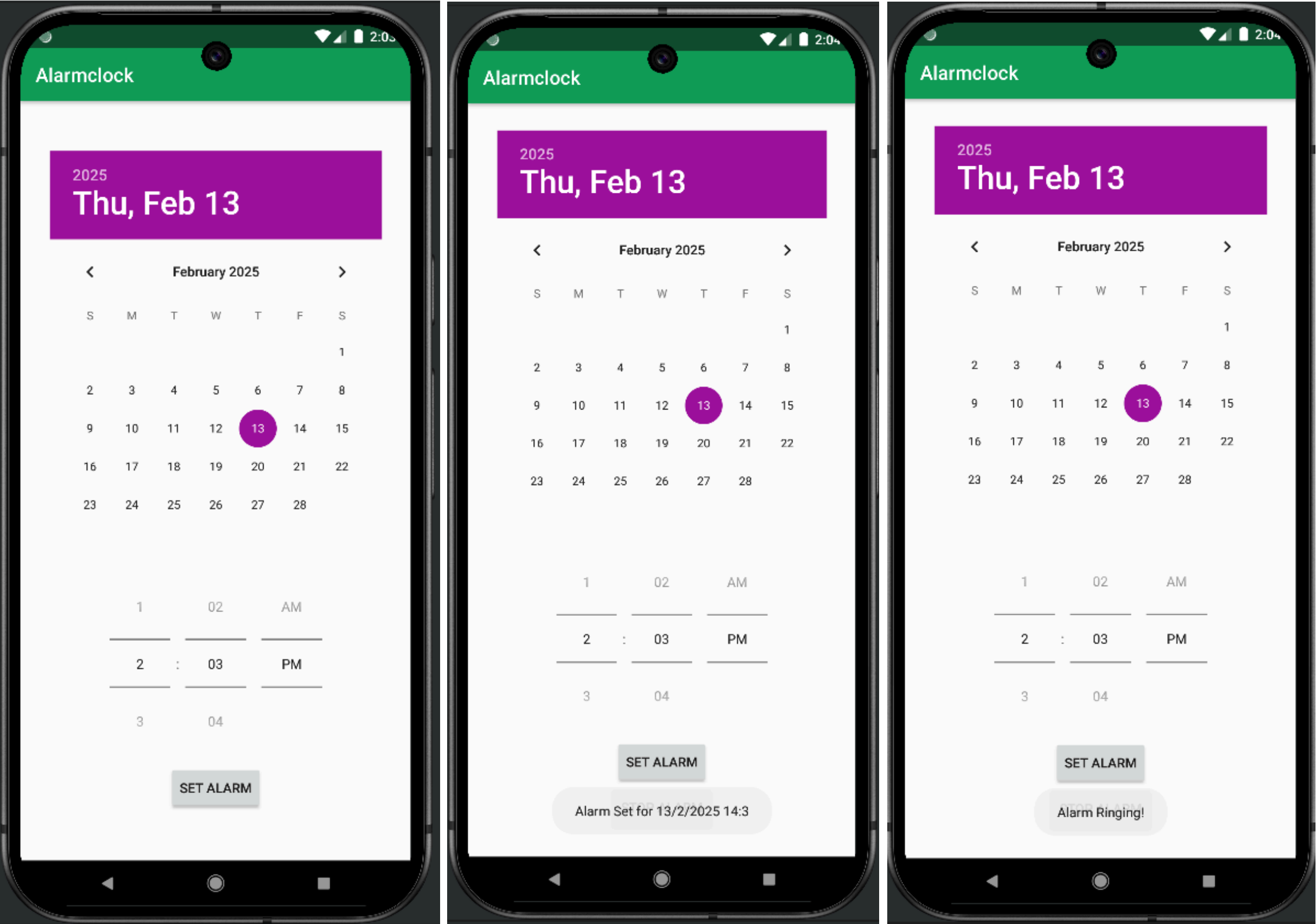
    @Override
    public void onReceive(Context context, Intent intent) {
        Toast.makeText(context, "Alarm Ringing!", Toast.LENGTH_LONG).show();

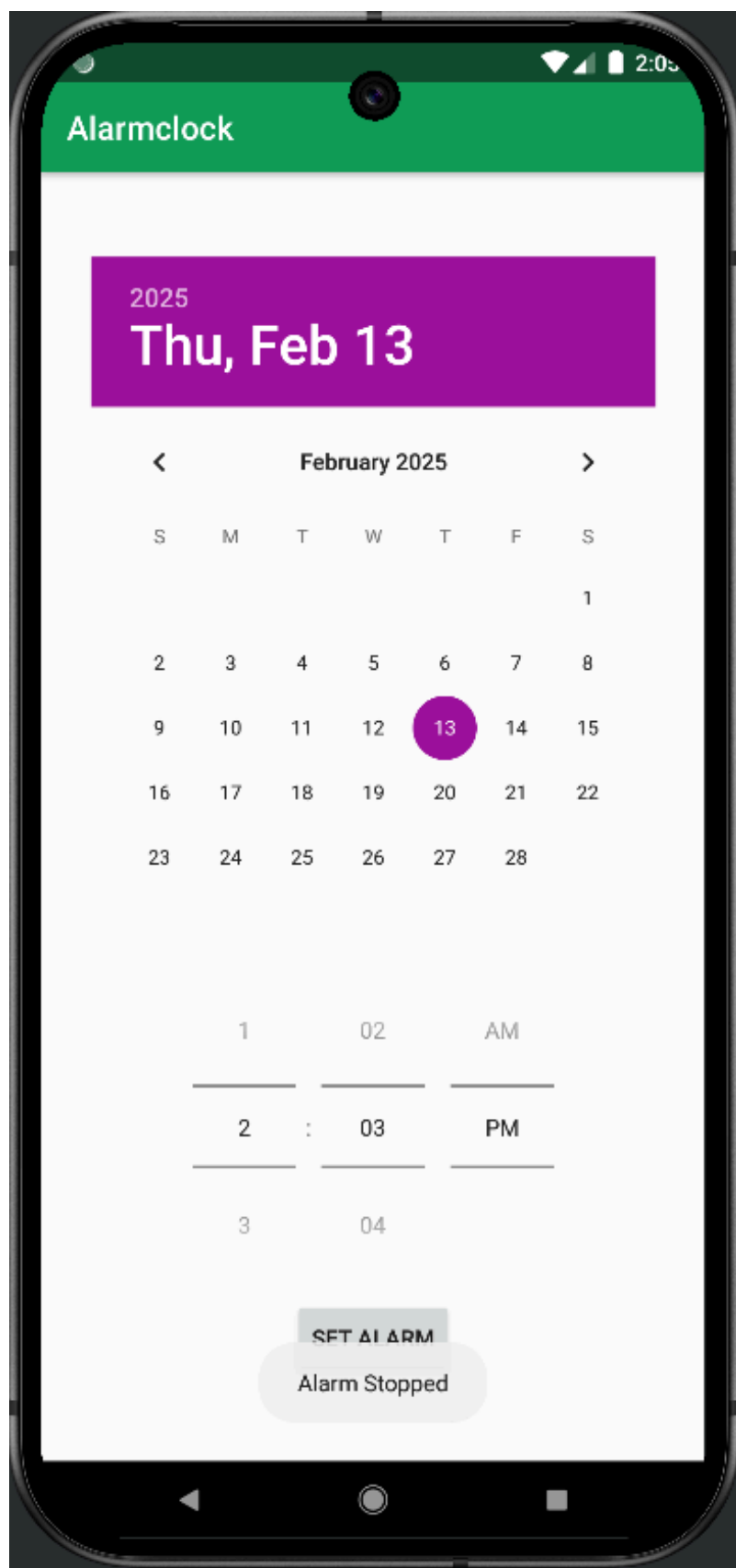
        if (ringtone == null) {
            Uri alarmUri = RingtoneManager.getDefaultUri(RingtoneManager.TYPE_ALARM);
            ringtone = RingtoneManager.getRingtone(context, alarmUri);
        }

        if (!ringtone.isPlaying()) {
            ringtone.play();
        }
    }

    public static void stopRingtone() {
        if (ringtone != null && ringtone.isPlaying()) {
            ringtone.stop();
            ringtone = null;
        }
    }
}
```

Output:

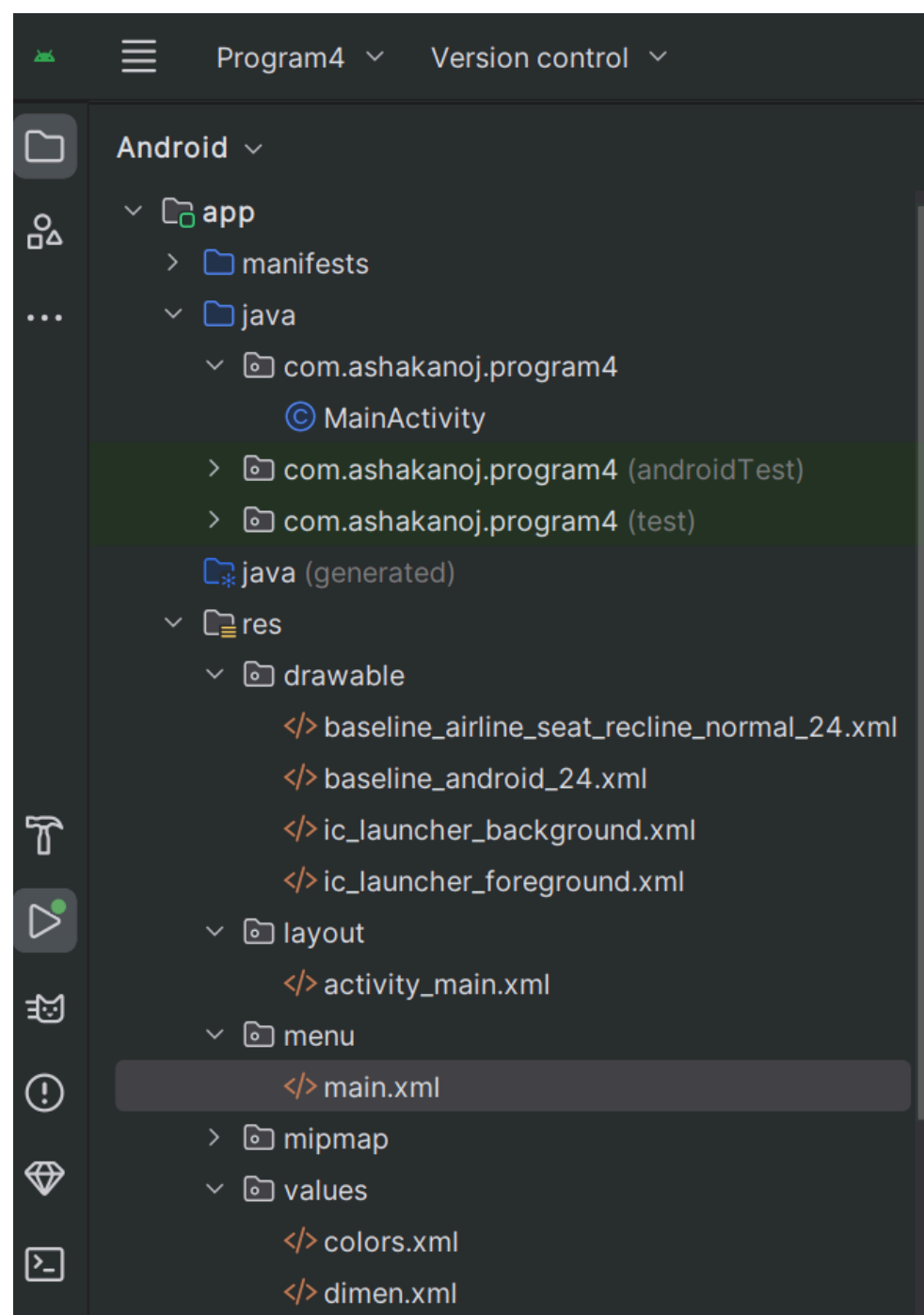




6. Create an UI with all views.

[Designing Android UI with Basic views | Android Programming by Wideskills](#)

STRUCTURE OF A PROGRAM:



Way To Create Menu Directory And Menu Resource File:

- To create the menu directory just right-click on **res** folder and navigate to **res->New->Android Resource Directory**.
- **Give resource directory name as menu and resource type also menu.** one directory will be created under res folder.
- To create xml resource file simply **right-click on menu folder** and navigate to **New->Menu Resource File**.
- Give name of file as menu_example. One menu_example.xml file will be created under menu folder.

activity_main.xml file:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="fill_parent"
    android:layout_height="fill_parent"
    android:orientation="vertical"
    android:paddingBottom="@dimen/activity_verticle_margin"
    android:paddingLeft="@dimen/activity_horizontal_margin"
    android:paddingRight="@dimen/activity_horizontal_margin"
    android:paddingTop="@dimen/activity_verticle_margin">

<TextView
    android:id="@+id/textView"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
```

```

        android:text="Hello Students!!" />

<Button
    android:id="@+id/buttonExample"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Example Button" />

<CheckBox
    android:id="@+id/checkboxExample"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="CheckBox Example" />

<ToggleButton
    android:id="@+id/toggleButtonExample"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="ToggleButton Example" />

<RadioGroup
    android:id="@+id/radioGroupExample"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content" >

    <RadioButton
        android:id="@+id/radioButton1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:checked="true"
        android:text="RadioButton 1" />

    <RadioButton
        android:id="@+id/radioButton2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="RadioButton 2" />

    <RadioButton
        android:id="@+id/radioButton3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="RadioButton 3" />
</RadioGroup>

<ImageButton
    android:id="@+id/imageButtonExample"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:src="@drawable/baseline_airline_seat_recline_normal_24" />

</LinearLayout>

```

Main.xml file:


```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

</menu>
```

dimen.xml file:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <dimen name="activity_verticle_margin">100dp</dimen>
    <dimen name="activity_horizontal_margin">100dp</dimen>
</resources>
```

MainActivity.java File:

```
package com.ashakanoj.program4;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.Menu;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.ImageButton;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.RadioGroup.OnCheckedChangeListener;
import android.widget.Toast;
import android.widget.ToggleButton;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Button example = (Button) findViewById(R.id.buttonExample);

        CheckBox exampleCheckBox = (CheckBox) findViewById(R.id.checkBoxExample);
        RadioGroup exampleRadioGroup = (RadioGroup) findViewById(R.id.radioGroupExample);
        ToggleButton exampleToggleButton = (ToggleButton)
        findViewById(R.id.toggleButtonExample);
        ImageButton exampleImageButton = (ImageButton)
        findViewById(R.id.imageButtonExample);

        example.setOnClickListener(new View.OnClickListener() {

            @Override
            public void onClick(View arg0) {
                // TODO Auto-generated method stub

                // Toast.makeText(this, "Hey Button is pressed !!",
                // Toast.LENGTH_SHORT).show();
                ToastToDisplay("Hey Button is pressed!!");
            }
        });
    }
}
```

```

    });
    exampleCheckBox.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            // TODO Auto-generated method stub
            if (((CheckBox) v).isChecked()) {
                ToastToDisplay("Check Box is checked");
            } else {
                ToastToDisplay("Check box is unchecked");
            }
        }
    });
    exampleRadioGroup
        .setOnCheckedChangeListener(new OnCheckedChangeListener() {
            @Override
            public void onCheckedChanged(RadioGroup group, int checkedId) {
                // TODO Auto-generated method stub
                RadioButton rb1 = (RadioButton) findViewById(R.id.radioBtton1);
                RadioButton rb2 = (RadioButton) findViewById(R.id.radioButton2);
                if (rb1.isChecked()) {
                    ToastToDisplay("Radio Button 1 is checked");
                } else if (rb2.isChecked()) {
                    ToastToDisplay("Radio Button 2 is checked");
                } else {
                    ToastToDisplay("Radio Button 3 is checked");
                }
            }
        });
    exampleToggleButton.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            // TODO Auto-generated method stub
            if (((ToggleButton) v).isChecked()) {
                ToastToDisplay("Toggle button is ON");
            } else {
                ToastToDisplay("Toggle Button is OFF");
            }
        }
    });

    exampleImageButton.setOnClickListener(new View.OnClickListener() {

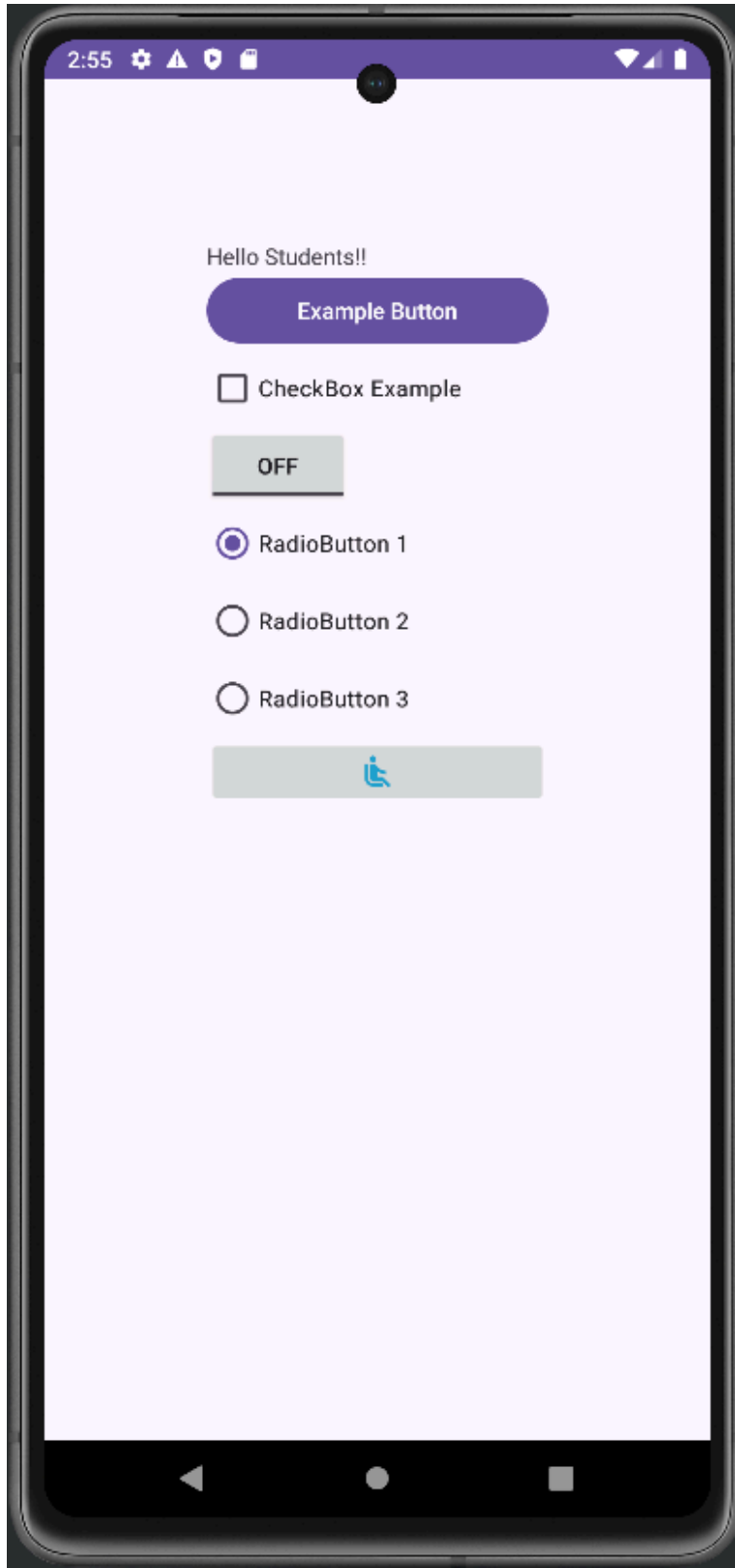
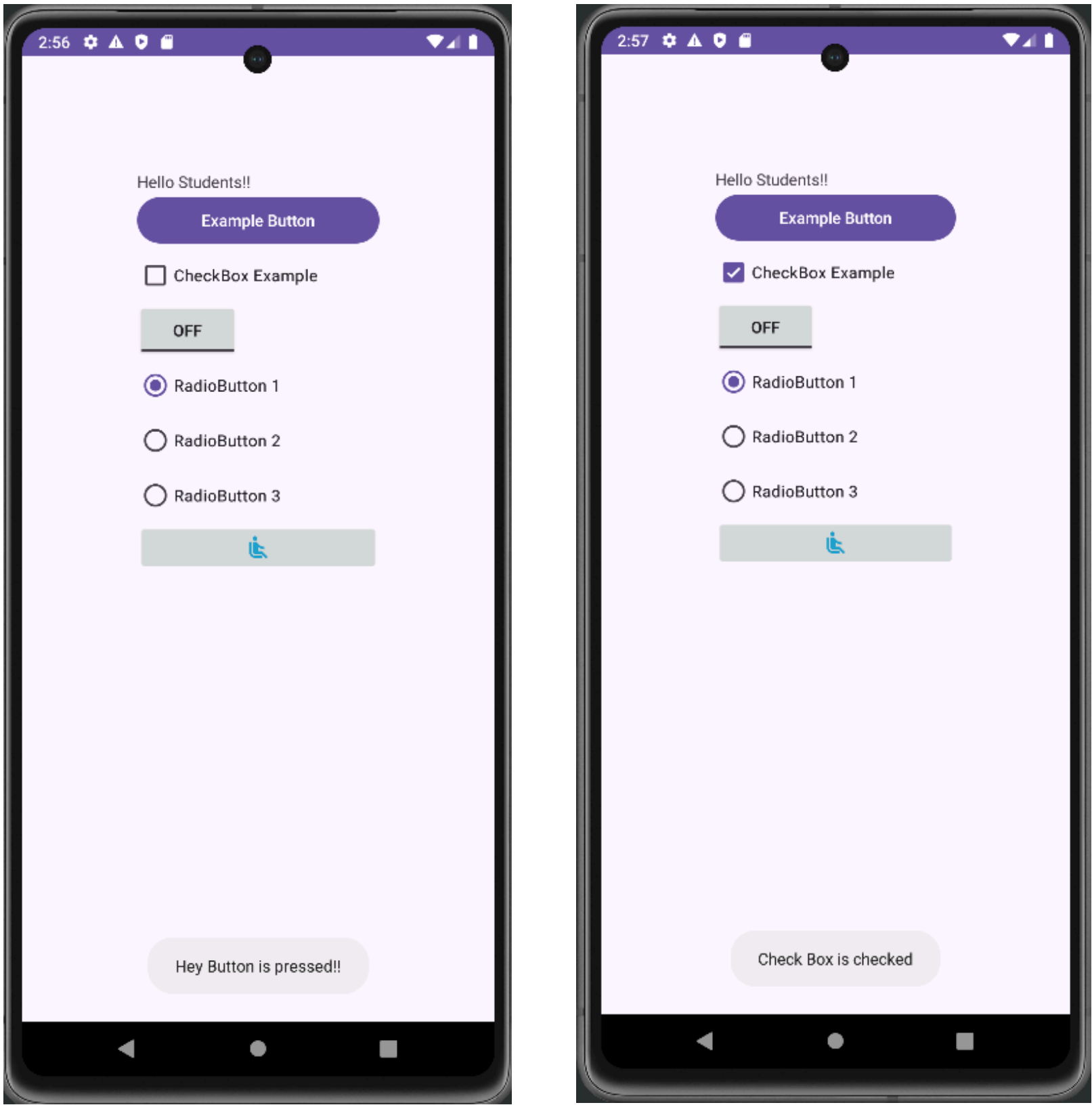
        @Override
        public void onClick(View v) {
            // TODO Auto-generated method stub
            ToastToDisplay("Image Button is pressed");
        }
    });
}

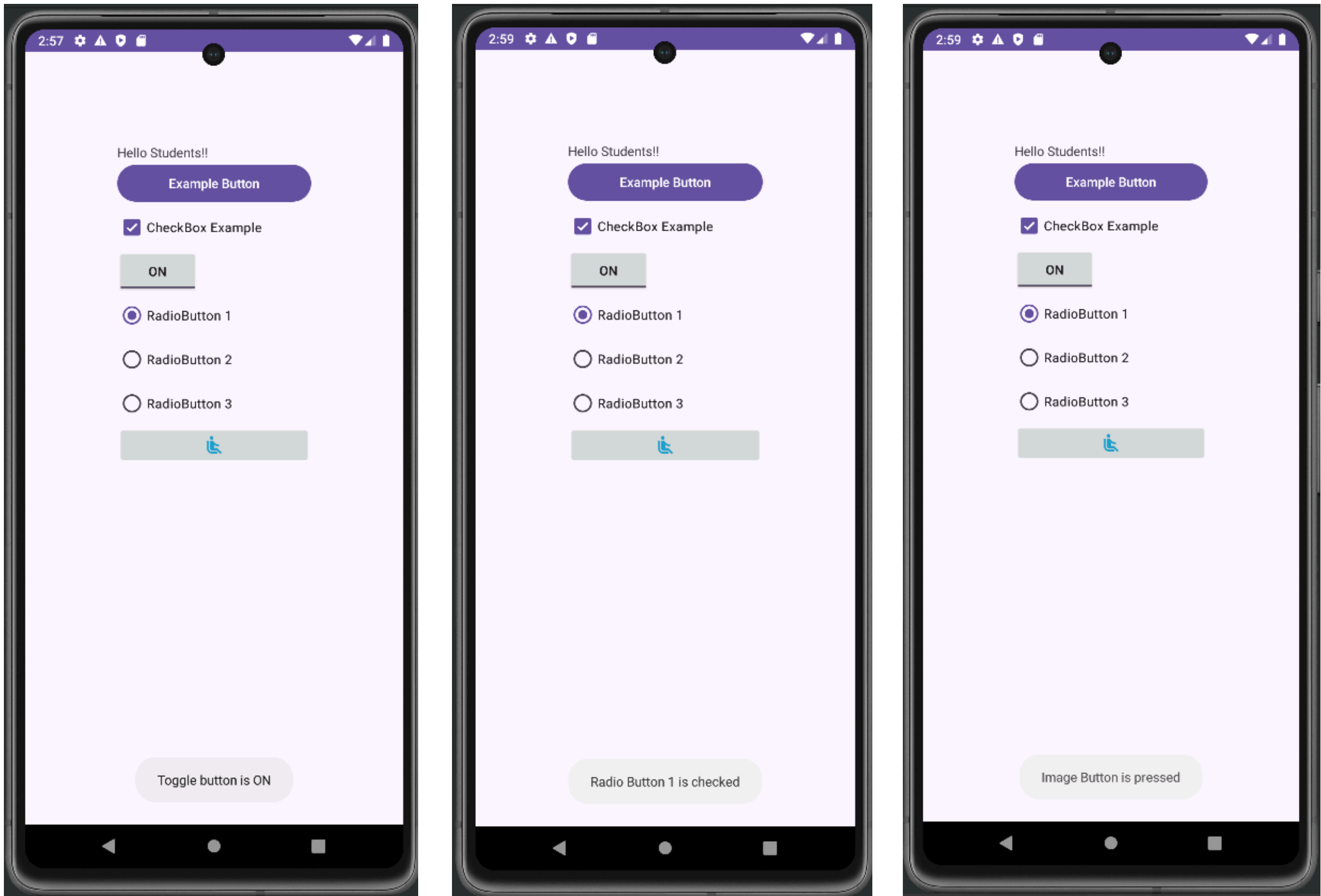
private void ToastToDisplay(String args) {
    Toast.makeText(getBaseContext(), args, Toast.LENGTH_SHORT).show();
}
@Override
public boolean onCreateOptionsMenu(Menu menu) {
    // Inflate the menu; this adds items to the action bar if it is present.
    getMenuInflater().inflate(R.menu.main, menu);
    return true;
}

```

```
}  
}
```

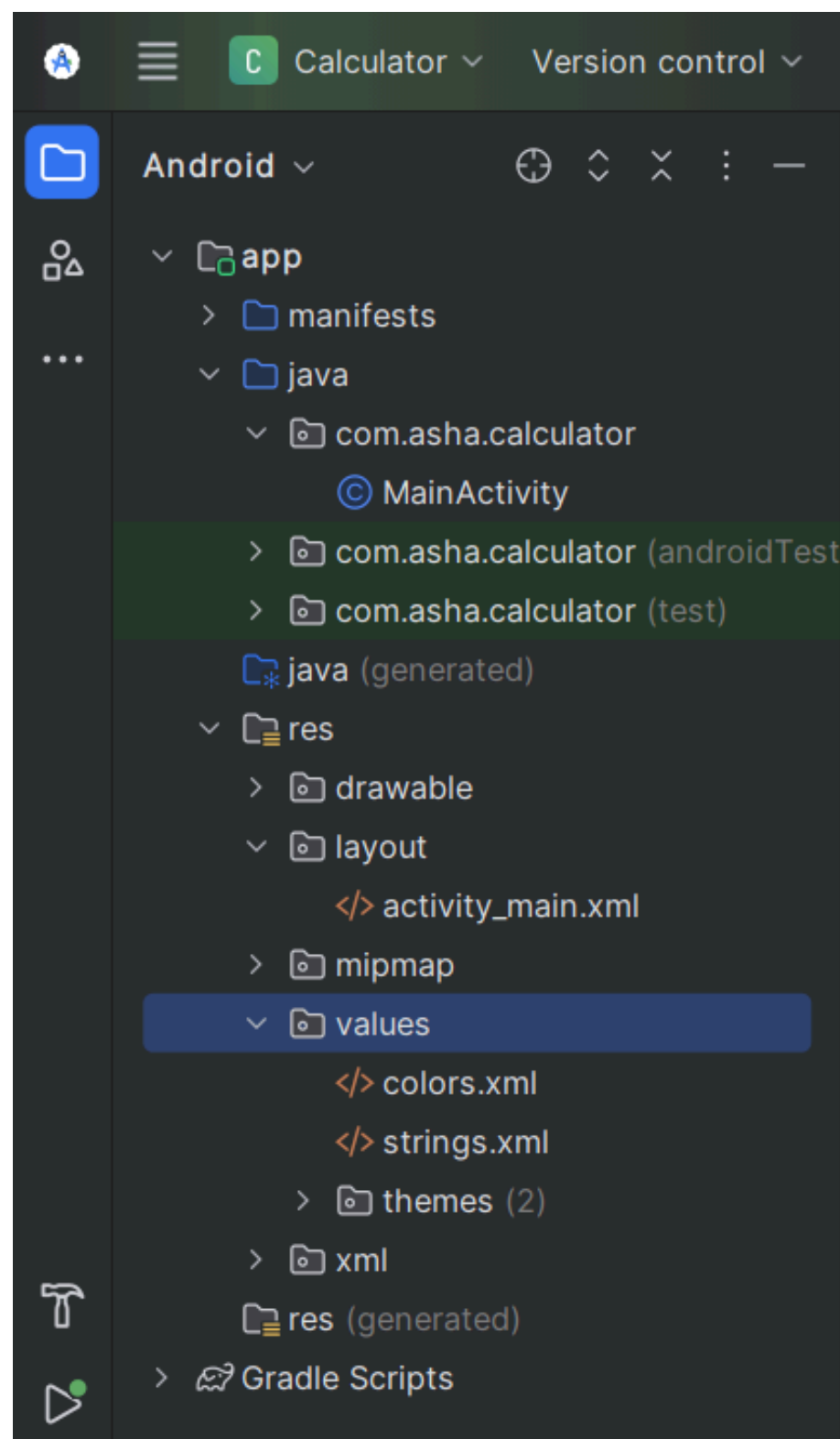
Output:





7. Create Simple Calculator in Android Application.

STRUCTURE OF A PROGRAM:



Activity_main.xml File:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:padding="16dp"
android:background="@color/aqua"
tools:context="MainActivity">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Simple Calculator"
        android:textColor="@color/black"
        android:textStyle="bold"
        android:textAlignment="center"
        android:textSize="30dp"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"/>

    <EditText
        android:id="@+id/num1EditText"
        android:layout_width="0dp"
```



```

        android:layout_height="48dp"
        android:layout_marginTop="44dp"
        android:hint="Enter number 1"
        android:inputType="numberDecimal"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

<EditText
    android:id="@+id/num2EditText"
    android:layout_width="0dp"
    android:layout_height="48dp"
    android:layout_marginTop="12dp"
    android:hint="Enter number 2"
    android:inputType="numberDecimal"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.47"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/num1EditText" />

<Button
    android:id="@+id/addButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:text="+"
    android:textSize="16sp"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/num2EditText" />

<Button
    android:id="@+id/subtractButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:text="-"
    android:textSize="16sp"
    app:layout_constraintEnd_toStartOf="@+id/multiplyButton"
    app:layout_constraintStart_toEndOf="@+id/addButton"
    app:layout_constraintTop_toBottomOf="@+id/num2EditText" />

<Button
    android:id="@+id/multiplyButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:text="x"
    android:textSize="16sp"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/num2EditText" />

<Button
    android:id="@+id/divideButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:text="/"
    android:textSize="16sp"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/addButton" />

```

```

<Button
    android:id="@+id/sqrtButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:layout_marginEnd="140dp"
    android:text="Sqrt"
    android:textSize="16sp"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/subtractButton" />

<TextView
    android:id="@+id/resultTextView"
    android:layout_width="84dp"
    android:layout_height="41dp"
    android:layout_marginStart="4dp"
    android:layout_marginTop="40dp"
    android:textStyle="bold"
    android:text="RESULT: "
    android:textSize="18sp"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/divideButton" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

MainActivity.java File:

```

package com.asha.calculator;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import com.asha.calculator.R;

import java.text.DecimalFormat;

public class MainActivity extends AppCompatActivity {

    // Declare variables to hold references to UI elements
    private EditText num1EditText, num2EditText;
    private TextView resultTextView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize UI elements from the layout
        num1EditText = findViewById(R.id.num1EditText);
        num2EditText = findViewById(R.id.num2EditText);
        resultTextView = findViewById(R.id.resultTextView);
    }
}

```

```

// Set click listeners for arithmetic operation buttons

Button addButton = findViewById(R.id.addButton);
addButton.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        performCalculation('+');
    }
});

Button subtractButton = findViewById(R.id.subtractButton);
subtractButton.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        performCalculation('-');
    }
});

Button multiplyButton = findViewById(R.id.multiplyButton);
multiplyButton.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        performCalculation('*');
    }
});

Button divideButton = findViewById(R.id.divideButton);
divideButton.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        performCalculation('/');
    }
});

Button sqrtButton = findViewById(R.id.sqrtButton);
sqrtButton.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        calculateSquareRoot();
    }
});
}

private void performCalculation(char operator) {
    // Get the values entered in the input fields
    String num1Str = num1EditText.getText().toString();
    String num2Str = num2EditText.getText().toString();

    // Check if either input field is empty
    if (num1Str.isEmpty() || num2Str.isEmpty()) {
        Toast.makeText(this, "Please enter both numbers", Toast.LENGTH_SHORT).show();
        return; // Exit the method to prevent calculations with empty inputs
    }

    // Convert the input values to numeric format
    double num1 = Double.parseDouble(num1Str);
    double num2 = Double.parseDouble(num2Str);
    double result = 0;

```



```

// Perform the selected calculation based on the operator
switch (operator) {
    case '+':
        result = num1 + num2;
        break;
    case '-':
        result = num1 - num2;
        break;
    case '*':
        result = num1 * num2;
        break;
    case '/':
        if (num2 != 0) {
            result = num1 / num2;
        } else {
            Toast.makeText(this, "Cannot divide by zero",
Toast.LENGTH_SHORT).show();
            return; // Exit the method if division by zero is attempted
        }
        break;
}

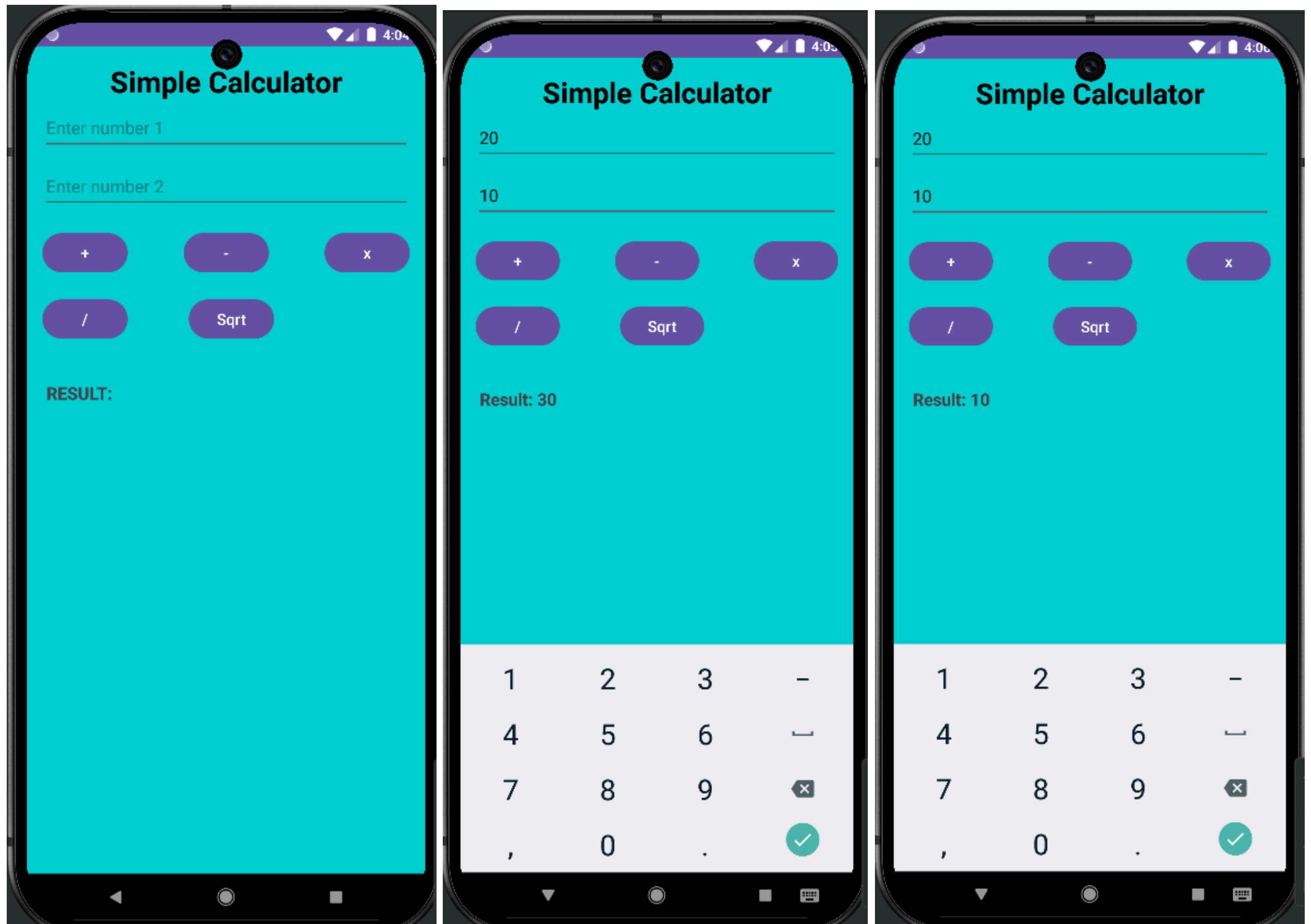
// Format and display the calculation result
DecimalFormat df = new DecimalFormat("#.##");
resultTextView.setText("Result: " + df.format(result));
}

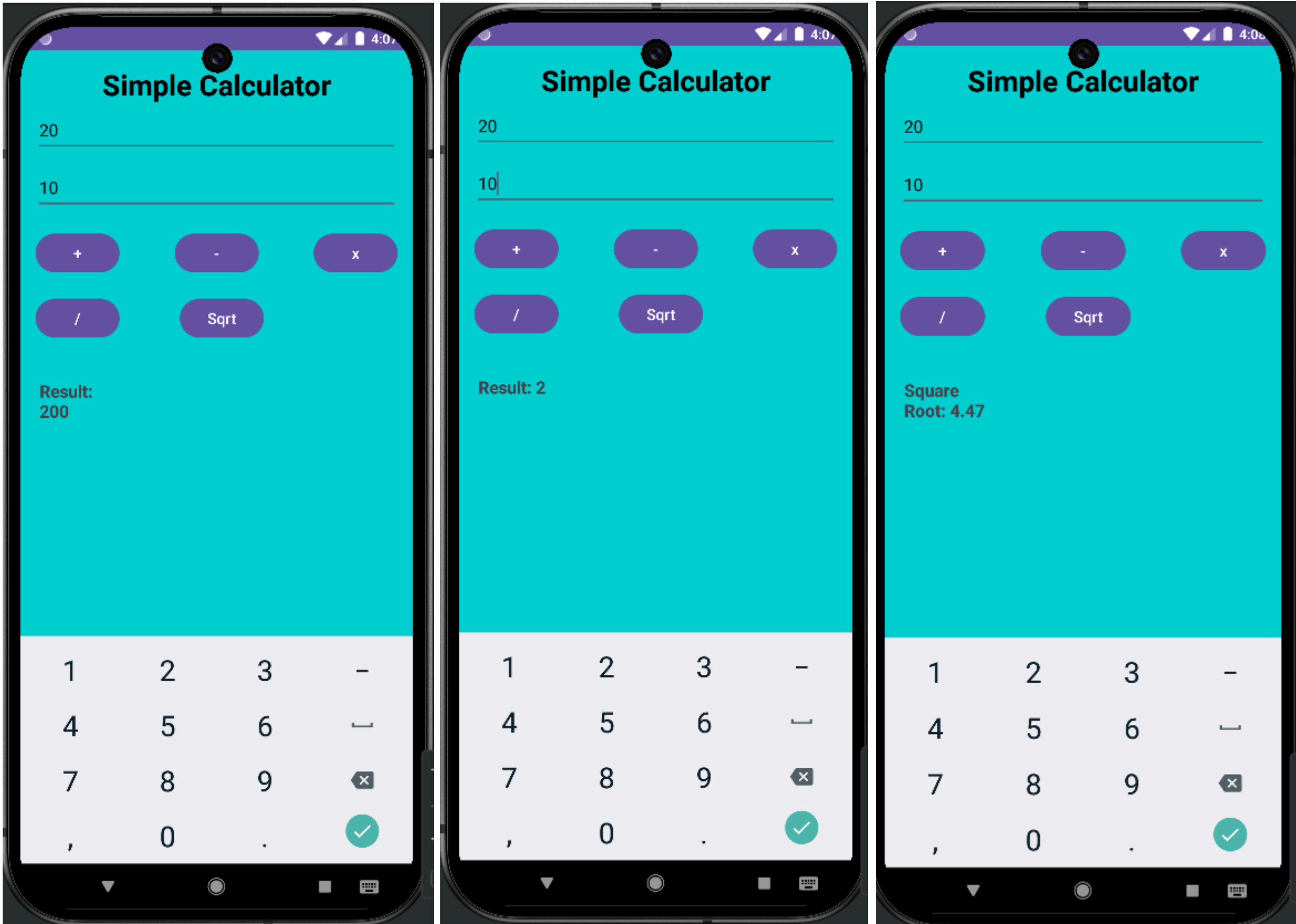
private void calculateSquareRoot() {
    String num1Str = num1EditText.getText().toString();
    // Check if the input field is empty
    if (num1Str.isEmpty()) {
        Toast.makeText(this, "Please enter a number", Toast.LENGTH_SHORT).show();
        return; // Exit the method to prevent calculations with empty inputs
    }

    double num = Double.parseDouble(num1Str);
    double sqrtResult = Math.sqrt(num);
    DecimalFormat df = new DecimalFormat("#.##");
    resultTextView.setText("Square Root: " + df.format(sqrtResult));
}
}

```


Output:

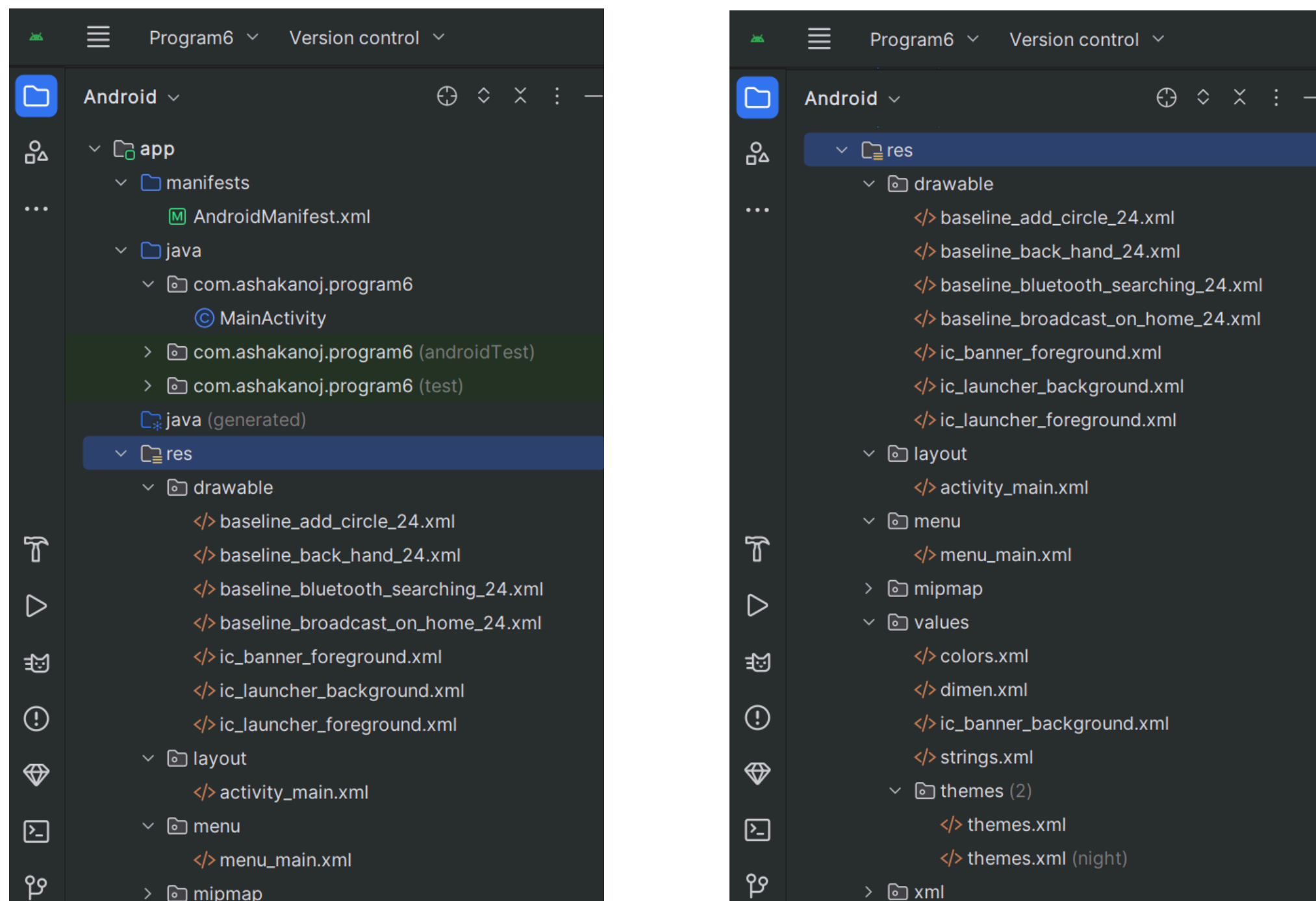




8. Create Menu in Application.

- ✓ <https://www.javatpoint.com/android-option-menu-example>
- ✓ <https://www.geeksforgeeks.org/how-to-implement-options-menu-in-android/>
- ✓ <https://developer.android.com/develop/ui/views/components/menus>
- ✓ <https://www.geeksforgeeks.org/how-to-implement-options-menu-in-android/>
- ✓ [Menu in Android | Android Development Tutorial | Studytonight](#)

STRUCTURE OF A PROGRAM:



Manifest.xml File:

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Program6"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
            </intent-filter>
        </activity>
    </application>
</manifest>
```

```

        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
</activity>
</application>

</manifest>

```

MainActivity.java File:

```

package com.ashakanoj.program6;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.ContextMenu;
import android.view.Menu;
import android.view.MenuInflater;
import android.view.MenuItem;
import android.view.View;
import android.widget.TextView;
import android.widget.Toast;
import android.widget.Toolbar;
import androidx.appcompat.widget.PopupMenu;

import static android.os.Build.VERSION_CODES.N;

public class MainActivity extends AppCompatActivity implements
    PopupMenu.OnMenuItemClickListener{
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        //Registering for Context Menu
        registerForContextMenu((TextView) findViewById(R.id.t1));
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        MenuInflater mi = getMenuInflater();
        mi.inflate(R.menu.menu_main, menu);
        return true;
    }

    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        int mid = item.getItemId();
        if (mid == R.id.i1) {
            Toast.makeText(this, "Clicked Option Menu",
                Toast.LENGTH_SHORT).show();
        } else if (mid == R.id.i2) {
            Toast.makeText(this, "I am Sub-menu 1",
                Toast.LENGTH_SHORT).show();
        } else if (mid == R.id.i3) {
            Toast.makeText(this, "I am Sub-menu 2",
                Toast.LENGTH_SHORT).show();
        }
        return super.onOptionsItemSelected(item);
    }

    @Override

```



```

public void onCreateContextMenu(ContextMenu menu, View v,
                                ContextMenu.ContextMenuInfo menuInfo) {
    super.onCreateContextMenu(menu, v, menuInfo);
    MenuInflater mi = getMenuInflater();
    mi.inflate(R.menu.menu_main, menu);
}
@Override
public boolean onOptionsItemSelected(MenuItem item) {
    int mid = item.getItemId();
    if (mid == R.id.i1) {
        Toast.makeText(this, "Clicked Context Menu",
            Toast.LENGTH_SHORT).show();
    } else if (mid == R.id.i2) {
        Toast.makeText(this, "I am Sub-menu 1",
            Toast.LENGTH_SHORT).show();
    } else if (mid == R.id.i3) {
        Toast.makeText(this, "I am Sub-menu 2",
            Toast.LENGTH_SHORT).show();
    }
    return super.onOptionsItemSelected(item);
}
public void pop(View v) {
    PopupMenu popup = new PopupMenu(this, v);
    MenuInflater mi = getMenuInflater();
    mi.inflate(R.menu.menu_main, popup.getMenu());
    popup.setOnMenuItemClickListener(this);
    popup.show();
}
@Override
public boolean onOptionsItemSelected(MenuItem item) {
    int mid = item.getItemId();
    if (mid == R.id.i1) {
        Toast.makeText(this, "Clicked Popup Menu",
            Toast.LENGTH_SHORT).show();
    } else if (mid == R.id.i2) {
        Toast.makeText(this, "I am Sub-menu 1",
            Toast.LENGTH_SHORT).show();
    } else if (mid == R.id.i3) {
        Toast.makeText(this, "I am Sub-menu 2",
            Toast.LENGTH_SHORT).show();
    }
    return true;
}
}
}

```

activity_main.xml File:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:paddingBottom="@dimen/activity_vertical_margin"
    android:paddingLeft="@dimen/activity_horizontal_margin"
    android:paddingRight="@dimen/activity_horizontal_margin"
    android:paddingTop="@dimen/activity_horizontal_margin"

```



```

    android:orientation="vertical"
    android:background="@color/skyblue"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/t1"
        android:layout_width="fill_parent"
        android:layout_height="50dp"
        android:text="I am Context Menu"
        android:textColor="@color/design_default_color_background"
        android:background="@color/yellowgreen"
        android:paddingBottom="20dp"
        android:backgroundTint="@color/design_default_color_secondary_variant"
        android:textAllCaps="true"
        android:textSize="20sp"
    />
    <Button
        android:id="@+id/button"
        android:layout_width="fill_parent"
        android:layout_height="50dp"
        android:onClick="pop"
        android:text="I am Pop Menu"
        android:textColor="@color/design_default_color_background"
        android:paddingBottom="20dp"
        android:background="@color/yellowgreen"
        android:backgroundTint="@color/design_default_color_secondary_variant"
        android:textAllCaps="true"
    />
</LinearLayout>

```

menu_main.xml File:

```

<?xml version="1.0" encoding="utf-8"?>
<menu
    xmlns:android="http://schemas.android.com/apk/res/android">
    <item android:id="@+id/i1"
        android:title="Click me to see Sub-menu Item"
        android:icon="@drawable/baseline_back_hand_24" >
        <!-- "item" submenu -->
        <menu>
            <item android:id="@+id/i2"
                android:title="Submenuitem 1"
                android:icon="@drawable/baseline_add_circle_24"
            />
            <item android:id="@+id/i3"
                android:title="Submenuitem 2"
                android:icon="@drawable/baseline_bluetooth_searching_24" />
        </menu>
    </item>
</menu>

```

Values: dimen.xml File:

```

<?xml version="1.0" encoding="utf-8"?>
<resources>
    <dimen name="activity_vertical_margin">100dp</dimen>
    <dimen name="activity_horizontal_margin">100dp</dimen>
</resources>

```


Values: colors.xml File:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
    <color name="green">#008000</color>
    <color name="yellowgreen">#9ACD32</color>
    <color name="blue">#0000FF</color>
    <color name="red">#FF0000</color>
    <color name="skyblue">#87CEEB</color>
</resources>
```

Values: string.xml File:

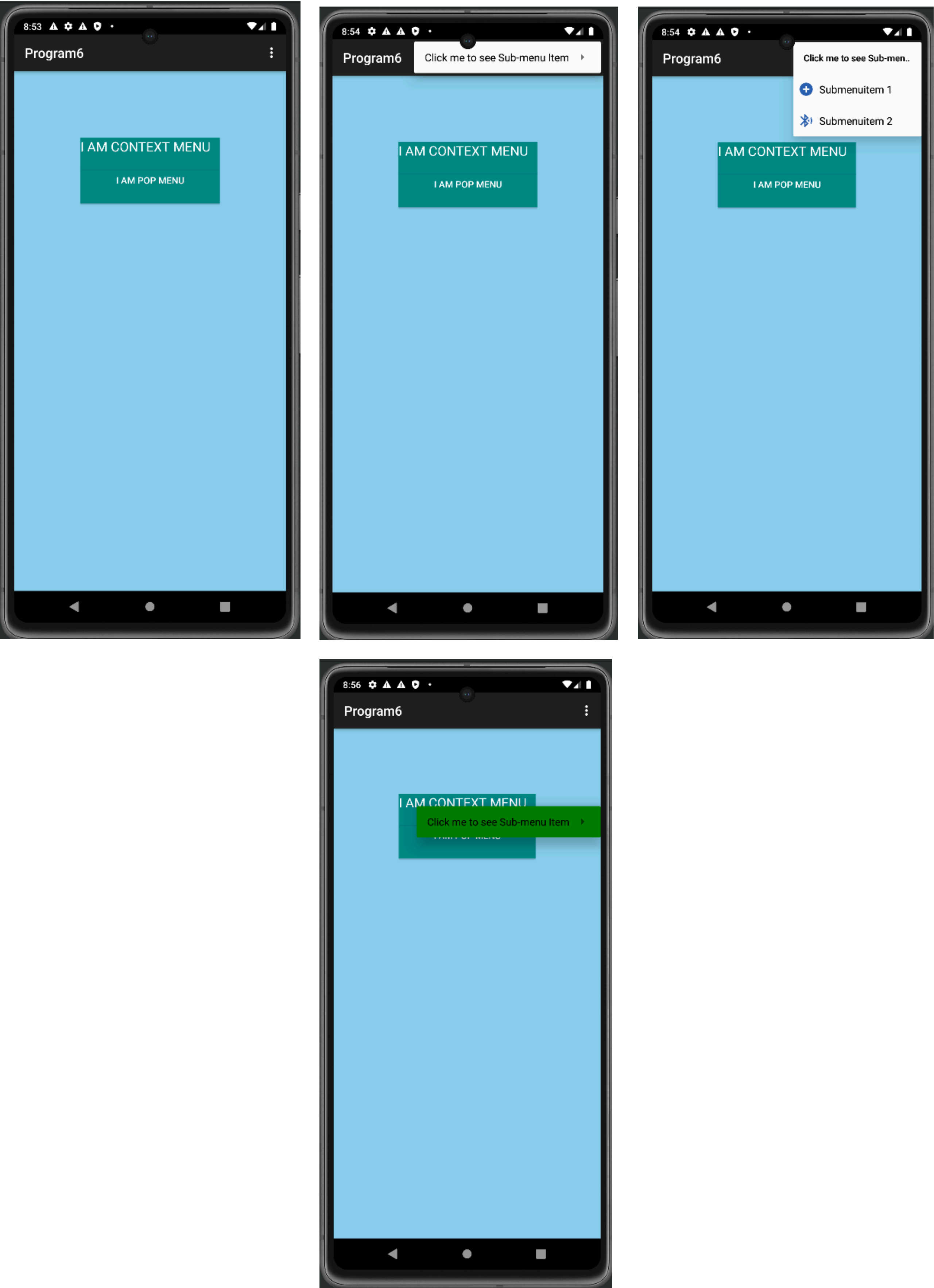
```
<resources>
    <string name="app_name">Program6</string>
    <!-- Base application theme. -->
    <style name="AppTheme" parent="Theme.AppCompat.Light.NoActionBar">
        <!-- Customize your theme here. -->
        <item name="colorPrimary">@color/design_default_color_primary_variant</item>
        <item name="colorPrimaryDark">@color/design_default_color_primary_dark</item>
        <item name="colorAccent">@color/design_default_color_secondary_variant</item>
    </style>
</resources>
```

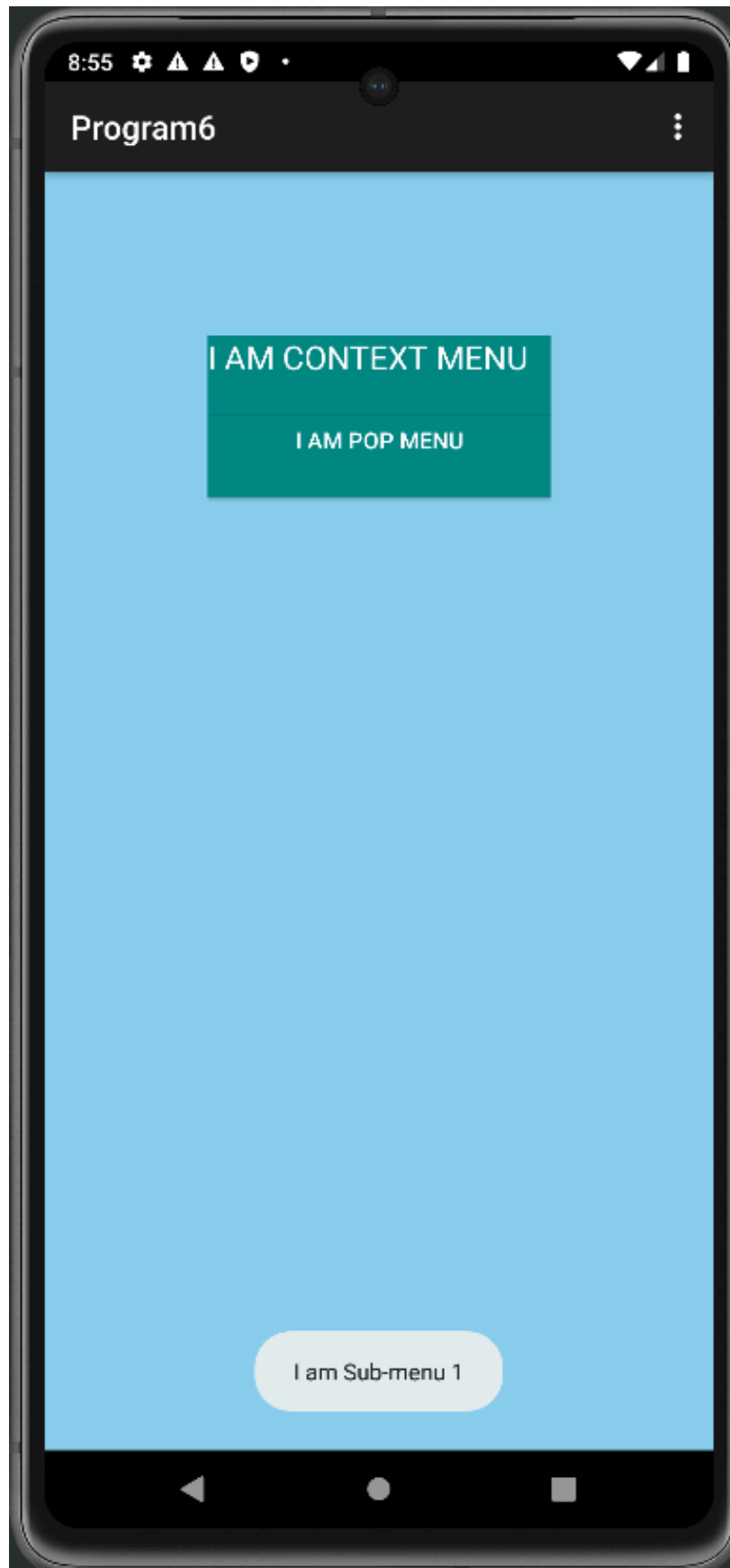
Themes: themes.xml File:

```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.Program6" parent="Theme.AppCompat.DayNight.DarkActionBar">
        <item name="android:colorBackground">@color/green</item>
        <item name="android:textColor">@color/black</item>
        <!-- Customize your light theme here. -->
        <!-- <item name="colorPrimary">@color/my_light_primary</item> -->
    </style>

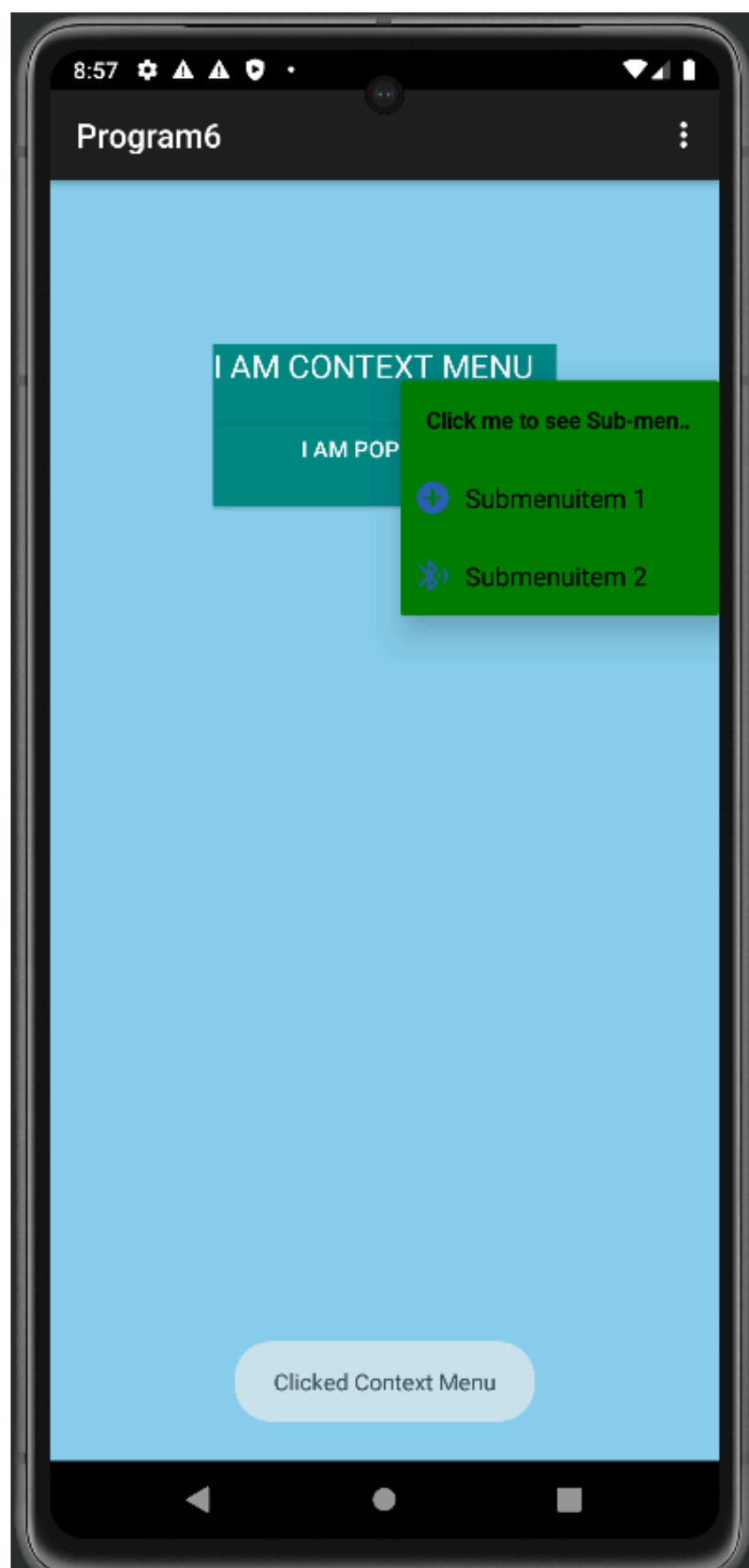
    <style name="Theme.Program6" parent="Base.Theme.Program6" />
</resources>
```

Output:



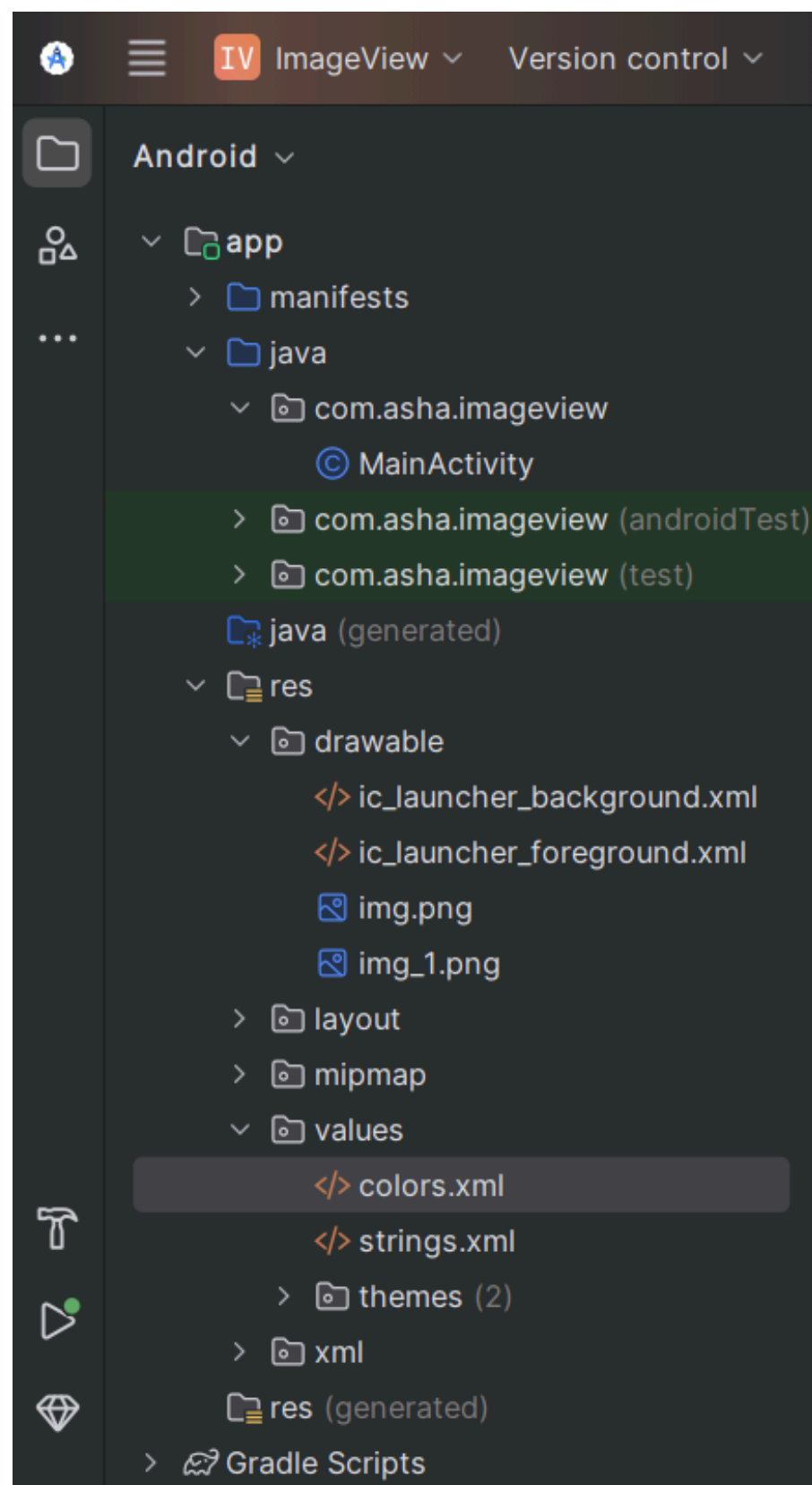


Long press on Context Menu to view the Option.



9. Create an application to display any two-image using Image View.

STRUCTURE OF A PROGRAM:



Activity_main.xml File:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp"
    android:paddingBottom="@dimen/activity_vertical_margin"
    android:paddingLeft="@dimen/activity_horizontal_margin"
    android:paddingRight="@dimen/activity_horizontal_margin"
    android:paddingTop="@dimen/activity_vertical_margin"
    android:background="#000">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click on Image"
        android:textSize="30dp"
        android:textColor="@color/white"
        android:textStyle="bold"/>

    <ImageView
        android:id="@+id/imageView1"
```

```

        android:layout_width="fill_parent"
        android:layout_height="250dp"
        android:src="@drawable/img"
        android:scaleType="centerCrop"
        android:layout_marginBottom="20dp"/>

<ImageView
    android:id="@+id/imageView2"
    android:layout_width="fill_parent"
    android:layout_height="250dp"
    android:src="@drawable/img_1"
    android:scaleType="centerCrop"/>
</LinearLayout>

```

MainActivity.java File:

```

package com.asha.imageview;

import android.os.Bundle;
import android.view.View;
import android.widget.ImageView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

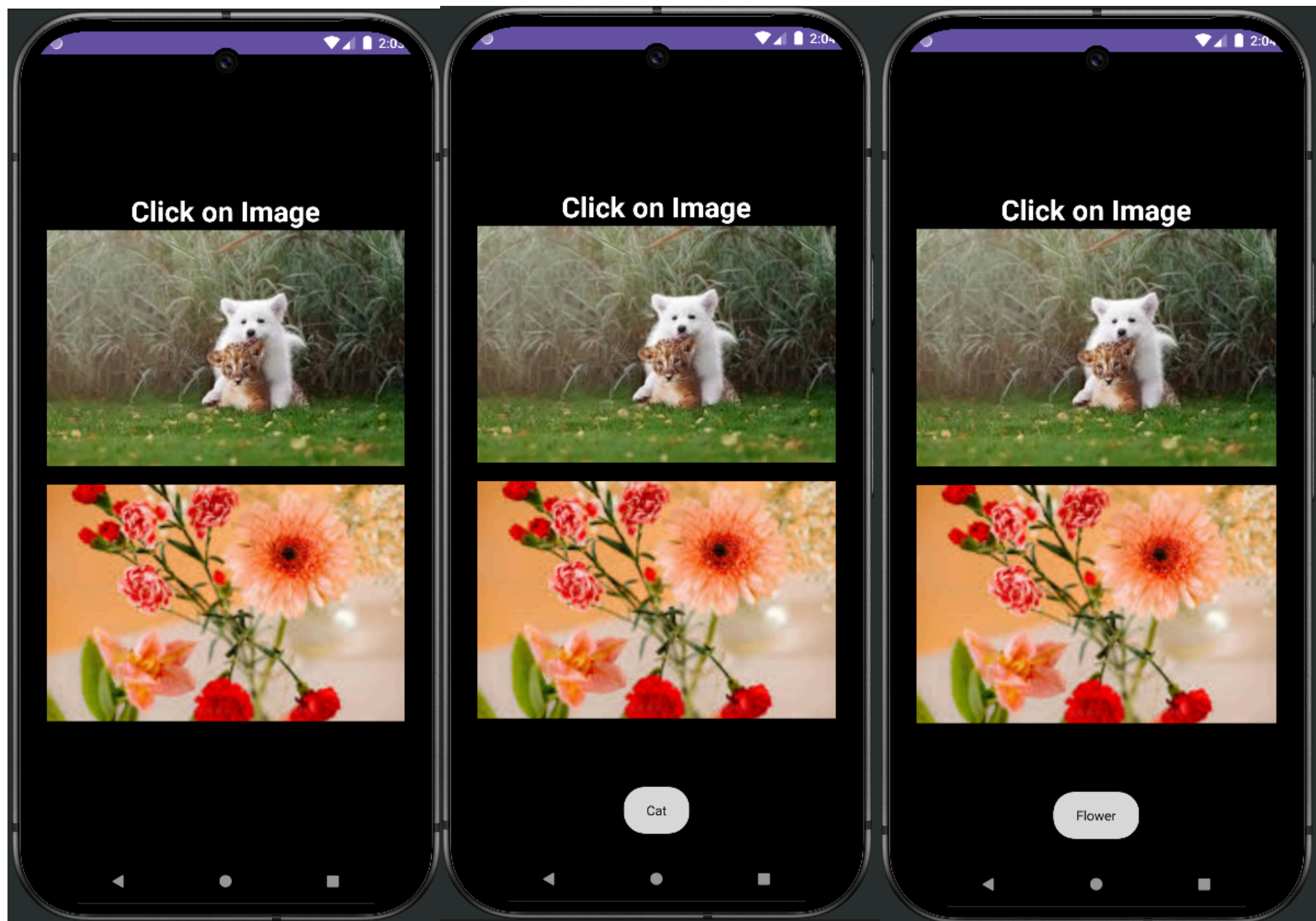
        // Reference ImageViews from the layout
        ImageView imageView1 = findViewById(R.id.imageView1);
        ImageView imageView2 = findViewById(R.id.imageView2);

        // Set images from drawable resources
        imageView1.setImageResource(R.drawable.img);
        imageView2.setImageResource(R.drawable.img_1);

        imageView1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Toast.makeText(getApplicationContext(),"Cat", Toast.LENGTH_LONG).show();
            }
        });
        imageView2.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Toast.makeText(getApplicationContext(),"Flower",
Toast.LENGTH_LONG).show();
            }
        });
    }
}

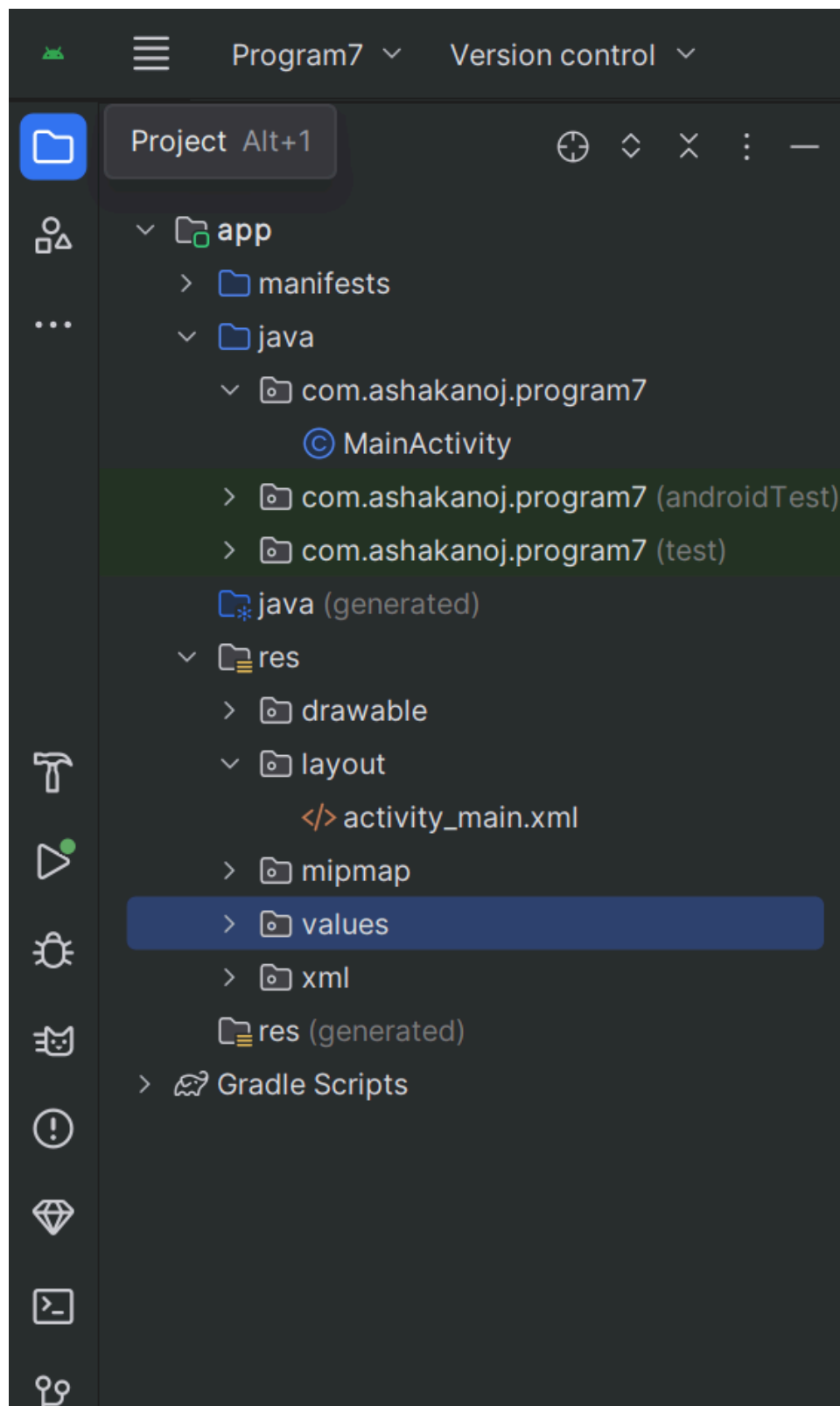
```


Output:



10. Read/ write the Local data.

https://www.tutorialspoint.com/android/android_internal_storage.htm
[Internal Storage Android - W3schools](https://www.w3schools.com/android/tryit.asp?filename=try_android_storage)

STRUCTURE OF A PROGRAM:**activity_main.xml File:**

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:paddingTop="@dimen/activity_vertical_margin"
    android:background="@color/black"
    tools:context=".MainActivity" >

    <EditText
        android:id="@+id/editText1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentTop="true"
        android:layout_alignParentRight="true"
        android:layout_marginTop="134dp"
        android:layout_marginRight="19dp"
        android:textColor="@color/white"
        android:ems="10">
```

```

        <requestFocus />
    </EditText>

    <EditText
        android:id="@+id/editText2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignRight="@+id/editText1"
        android:layout_below="@+id/editText1"
        android:layout_marginTop="24dp"
        android:textColor="@color/white"
        android:ems="10" />

    <TextView
        android:id="@+id/textView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignBaseline="@+id/editText1"
        android:layout_alignBottom="@+id/editText1"
        android:layout_alignParentLeft="true"
        android:textColor="@color/white"
        android:text="Enter File Name:" />

    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignBaseline="@+id/editText2"
        android:layout_alignBottom="@+id/editText2"
        android:layout_alignParentLeft="true"
        android:textColor="@color/white"
        android:text=" Enter Information to Read:" />

    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignLeft="@+id/editText2"
        android:layout_below="@+id/editText2"
        android:layout_marginLeft="50dp"
        android:layout_marginTop="16dp"
        android:text="Save" />

    <Button
        android:id="@+id/button2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignBaseline="@+id/button1"
        android:layout_alignBottom="@+id/button1"
        android:layout_toRightOf="@+id/button1"
        android:text="Read" />

</RelativeLayout>

```


MainActivity.xml File:

```

package com.ashakanoj.program7;
import android.content.Context;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import java.io.BufferedReader;
import java.io.FileNotFoundException;
import java.io.FileOutputStream;
import java.io.IOException;
import java.io.InputStreamReader;

public class MainActivity extends AppCompatActivity {
    EditText editTextFileName,editTextData;
    Button saveButton,readButton;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        editTextFileName=findViewById(R.id.editText1);
        editTextData=findViewById(R.id.editText2);
        saveButton=findViewById(R.id.button1);
        readButton=findViewById(R.id.button2);

        //Performing Action on Read Button
        saveButton.setOnClickListener(new View.OnClickListener(){

            @Override
            public void onClick(View arg0) {
                String filename=editTextFileName.getText().toString();
                String data=editTextData.getText().toString();

                FileOutputStream fos;
                try {
                    fos = openFileOutput(filename, Context.MODE_PRIVATE);
                    //default mode is PRIVATE, can be APPEND etc.
                    fos.write(data.getBytes());
                    fos.close();

                    Toast.makeText(getApplicationContext(),filename + " stored",
                        Toast.LENGTH_LONG).show();

                } catch (FileNotFoundException e) {e.printStackTrace();}
                catch (IOException e) {e.printStackTrace();}

            }

        });
    }
}

```

```

//Performing Action on Read Button
readButton.setOnClickListener(new View.OnClickListener(){

    @Override
    public void onClick(View arg0) {
        String filename=editTextFileName.getText().toString();
        StringBuffer stringBuffer = new StringBuffer();
        try {
            //Attaching BufferedReader to the FileInputStream by the help of
InputStreamReader
            BufferedReader inputReader = new BufferedReader(new InputStreamReader(
                openFileInput(filename)));
            String inputString;
            //Reading data line by line and storing it into the stringbuffer
            while ((inputString = inputReader.readLine()) != null) {
                stringBuffer.append(inputString + "\n");
            }

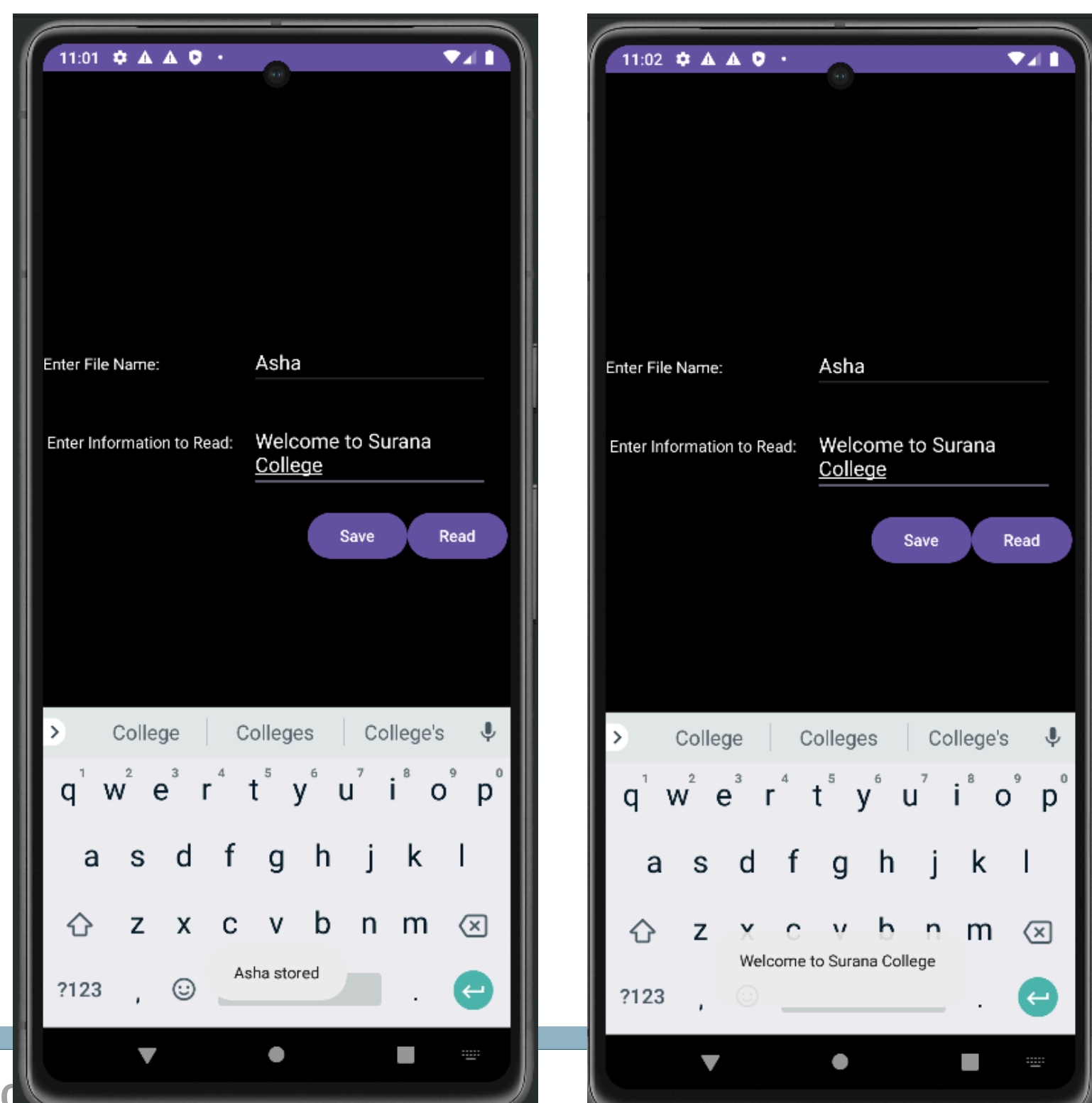
        } catch (IOException e) {
            e.printStackTrace();
        }
        //Displaying data on the toast
Toast.makeText(getApplicationContext(),stringBuffer.toString(),Toast.LENGTH_LONG).show();

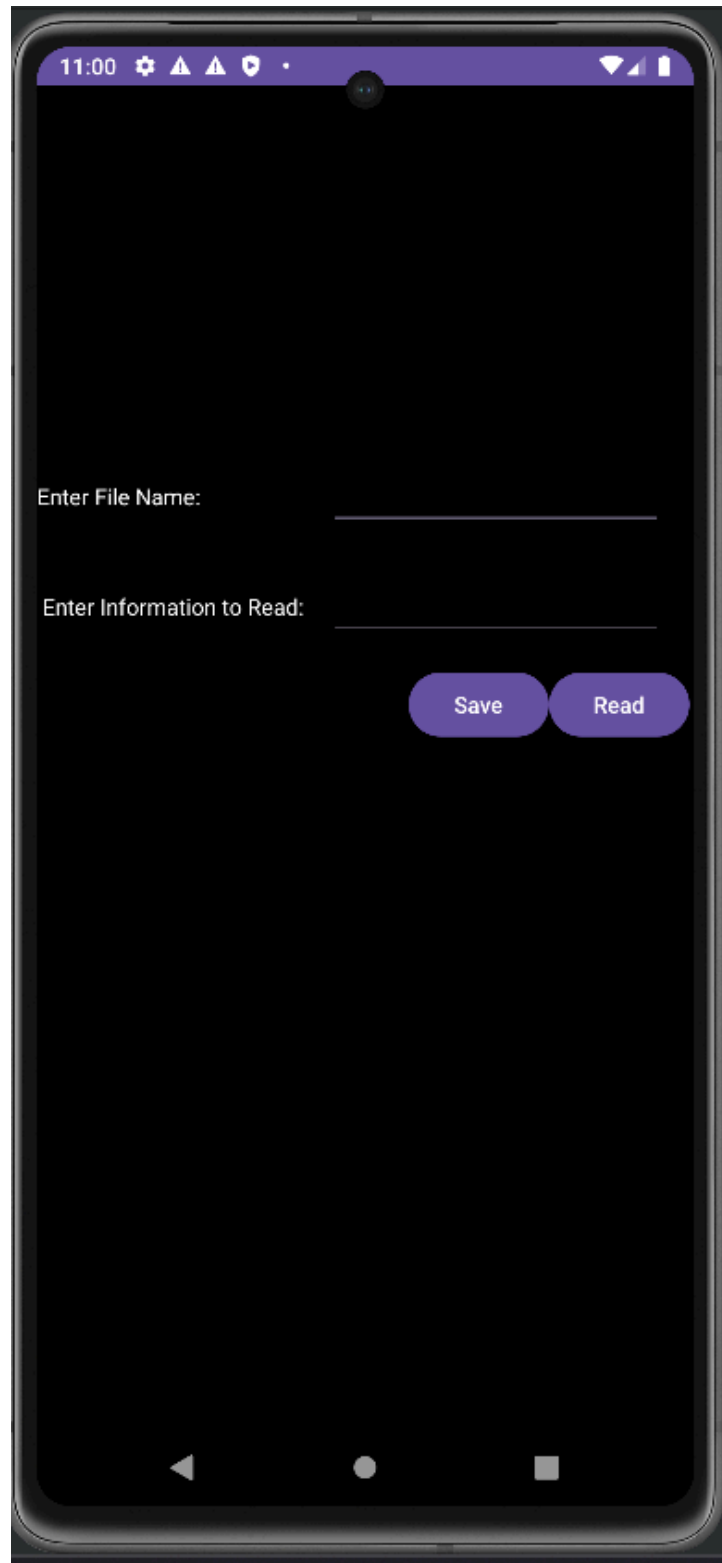
    }

});
}
}

```

Output:





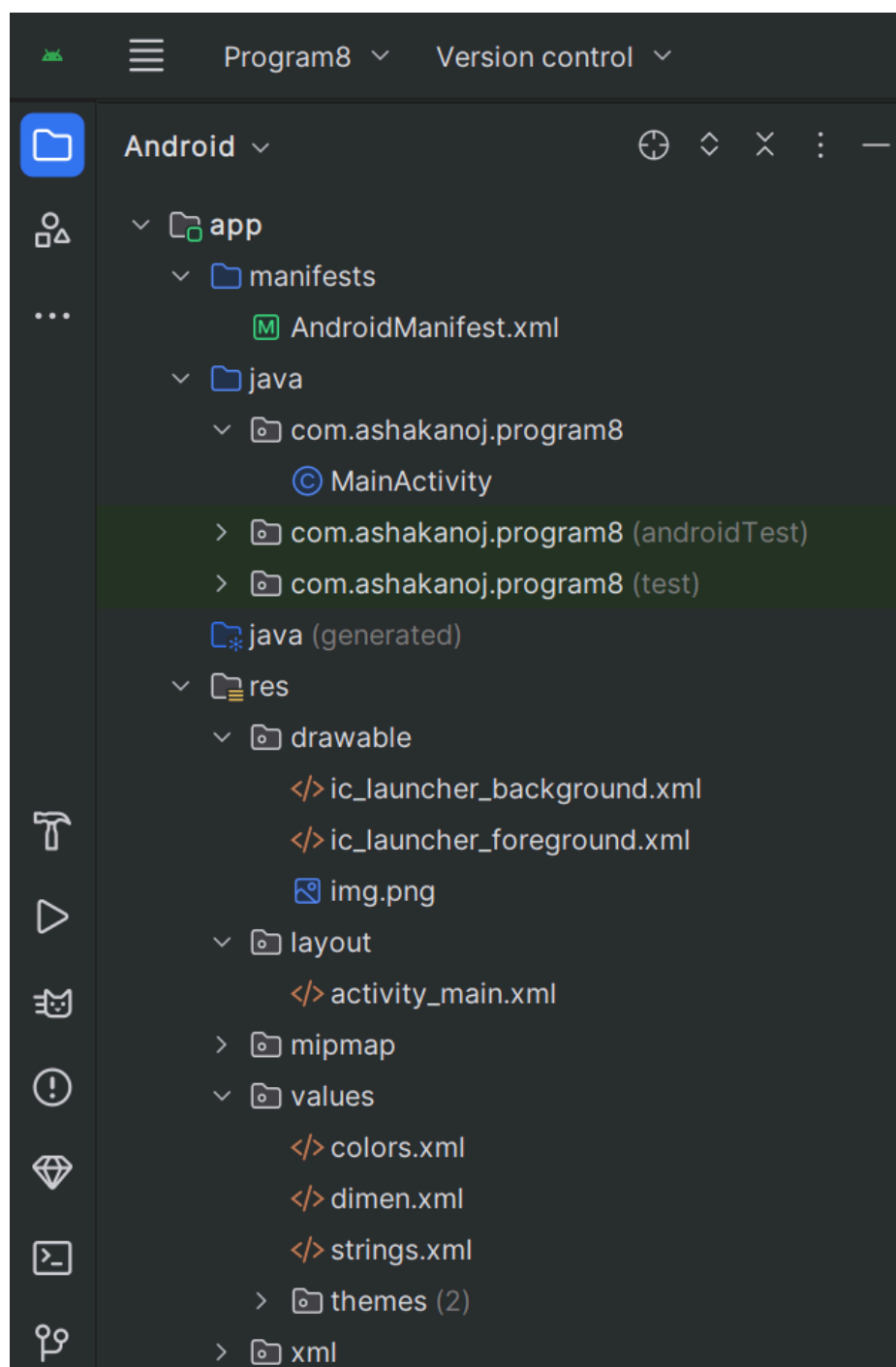
11. Create an application to send SMS.

https://www.tutorialspoint.com/android/android_sending_sms.htm

<https://www.geeksforgeeks.org/sending-a-text-message-over-the-phone-using-smsmanager-in-android/>

<https://www.geeksforgeeks.org/implement-phone-number-validator-in-android/>

STRUCTURE OF A PROGRAM:



Androidmanifest.xml File:

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <uses-permission android:name="android.permission.SEND_SMS"/>
    <uses-feature android:name="android.hardware.telephony" android:required="false" />
    <uses-permission android:name="android.permission.READ_PHONE_STATE"/>

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.SMS"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>
```

Activity_main.xml File:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp"
    android:background="@color/pink"
    android:orientation="vertical">

    <EditText
        android:id="@+id/phoneNumber"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Phone Number"
        android:inputType="phone"/>

    <EditText
        android:id="@+id/message"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Message"
        android:inputType="textMultiLine"
        android:minHeight="100dp"/>

    <Button
        android:id="@+id/sendSMS"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Send SMS"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

MainActivity.java File:

```
package com.asha.sms;
import android.Manifest;
import android.content.pm.PackageManager;
import android.os.Bundle;
import android.telephony.SmsManager;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;

public class MainActivity extends AppCompatActivity {

    EditText phoneNumber, message;
    Button sendSMS;
    private static final int SMS_PERMISSION_CODE = 101;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```



```

setContentView(R.layout.activity_main);

// Initialize UI components
phoneNumber = findViewById(R.id.phoneNumber);
message = findViewById(R.id.message);
sendSMS = findViewById(R.id.sendSMS);

// Button click listener
sendSMS.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        if (checkSmsPermission()) {
            sendSms();
        } else {
            requestSmsPermission();
        }
    }
});

// Method to send SMS with validation
private void sendSms() {
    String number = phoneNumber.getText().toString().trim();
    String msg = message.getText().toString().trim();

    // Validate phone number
    if (number.isEmpty()) {
        Toast.makeText(this, "Please enter a phone number",
Toast.LENGTH_SHORT).show();
        return;
    }

    if (!number.matches("\\d{10}")) {
        Toast.makeText(this, "Enter a valid 10-digit phone number",
Toast.LENGTH_SHORT).show();
        return;
    }

    // Validate message content
    if (msg.isEmpty()) {
        Toast.makeText(this, "Please enter a message", Toast.LENGTH_SHORT).show();
        return;
    }

    try {
        SmsManager smsManager = SmsManager.getDefault();
        smsManager.sendTextMessage(number, null, msg, null, null);
        Toast.makeText(this, "SMS Sent Successfully!", Toast.LENGTH_SHORT).show();
    } catch (Exception e) {
        Toast.makeText(this, "SMS Sending Failed!", Toast.LENGTH_SHORT).show();
        e.printStackTrace();
    }
}

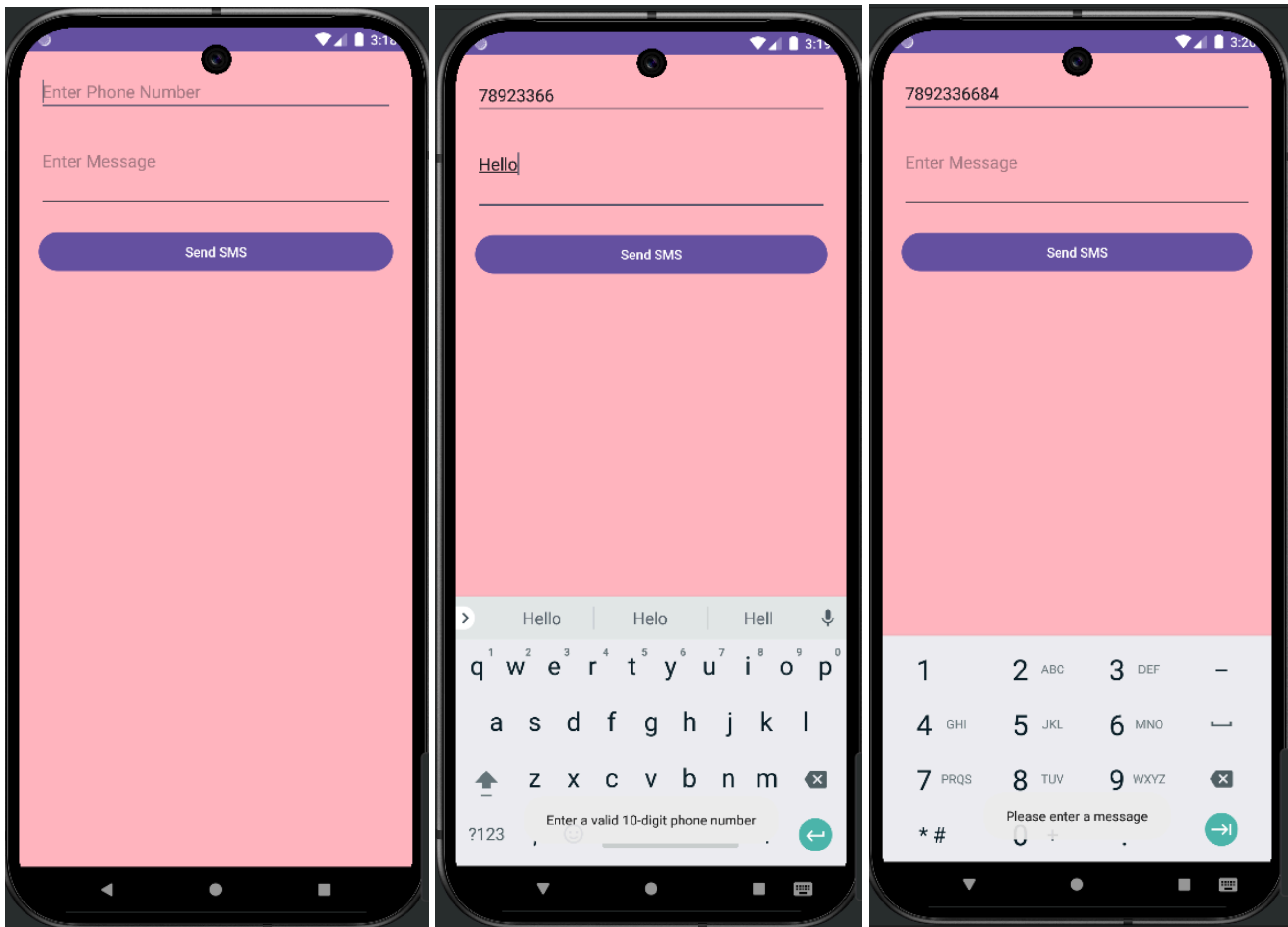
// Check SMS permission
private boolean checkSmsPermission() {
    return ContextCompat.checkSelfPermission(this, Manifest.permission.SEND_SMS) ==
PackageManager.PERMISSION_GRANTED;
}

```

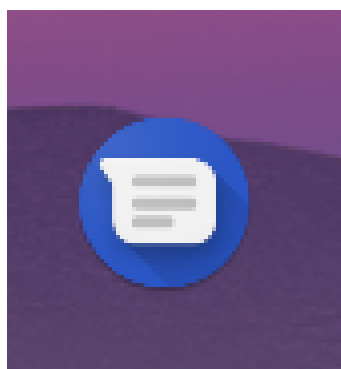


```
// Request SMS permission
private void requestSmsPermission() {
    ActivityCompat.requestPermissions(this, new String[]{Manifest.permission.SEND_SMS},
SMS_PERMISSION_CODE);
}

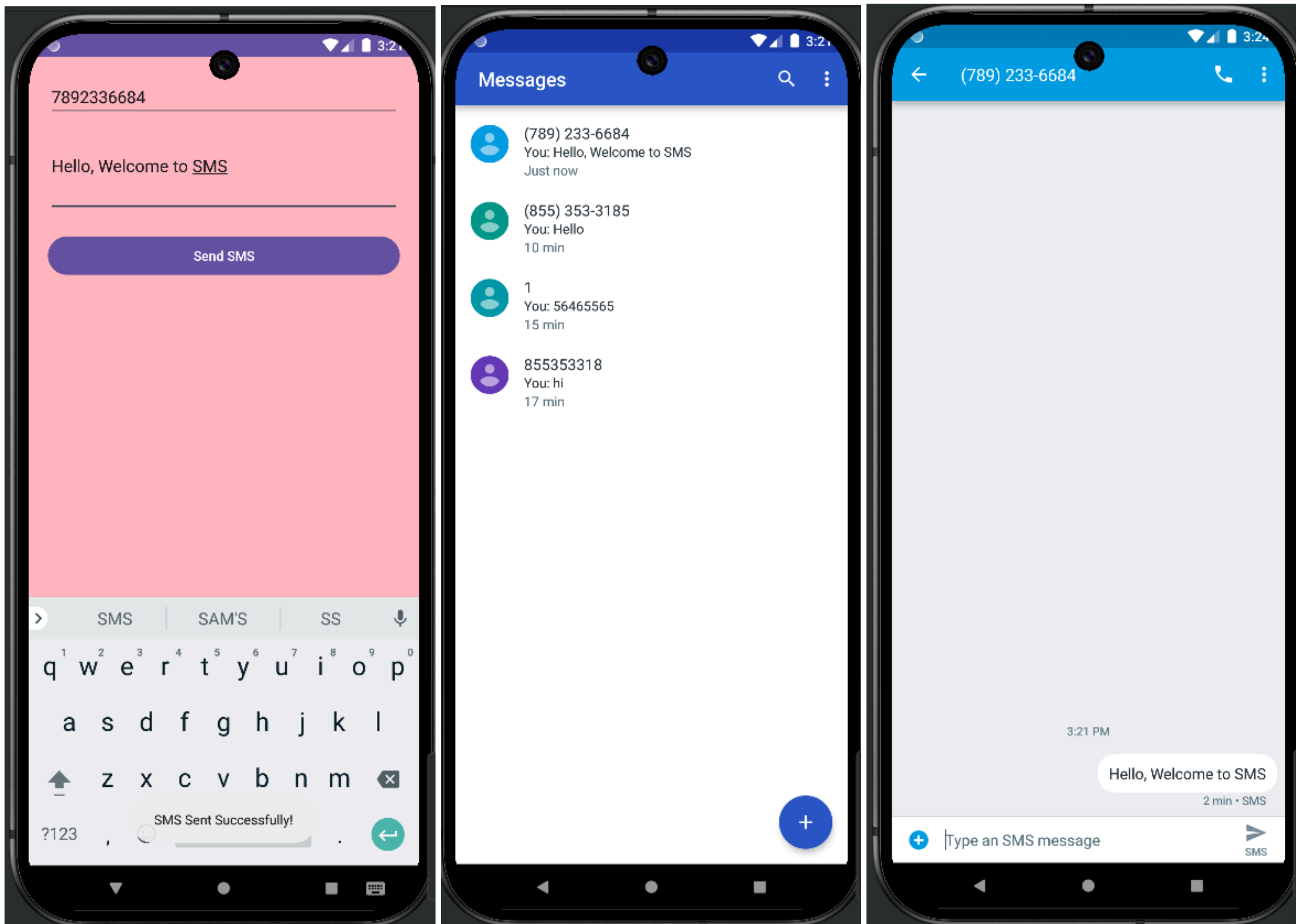
// Handle permission result
@Override
public void onRequestPermissionsResult(int requestCode, @NonNull String[] permissions,
@NonNull int[] grantResults) {
    super.onRequestPermissionsResult(requestCode, permissions, grantResults);
    if (requestCode == SMS_PERMISSION_CODE) {
        if (grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
            sendSms();
        } else {
            Toast.makeText(this, "Permission Denied! Cannot send SMS.",
Toast.LENGTH_SHORT).show();
        }
    }
}
}
```

Output:

Go to SMS Widget in the Virtual Device and click on Icon given below:

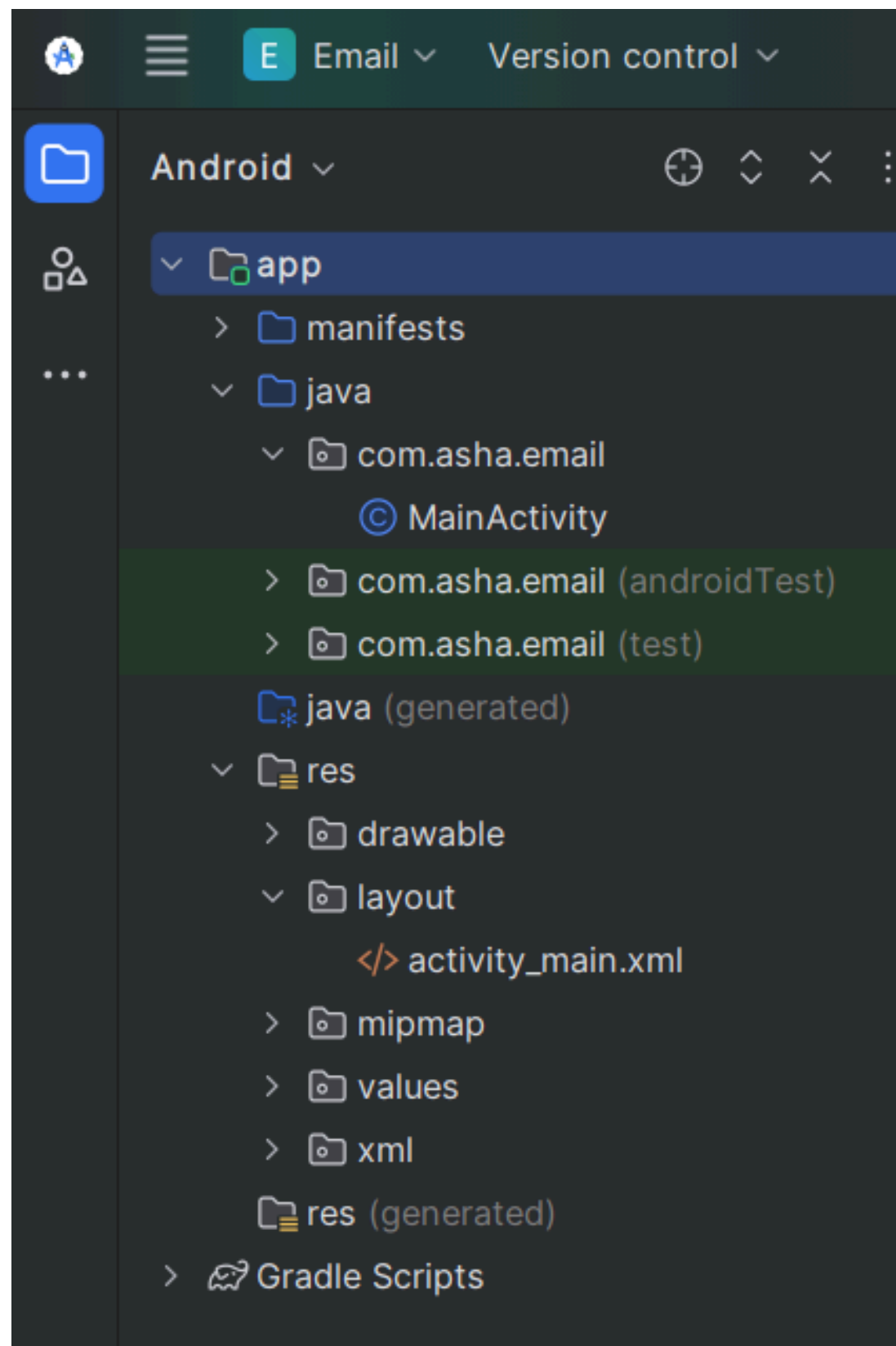


You can view the received SMS as shown below.



12. Create an application to send an e-mail.

STRUCTURE OF A PROGRAM:



activity_main.xml File:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp"
    android:orientation="vertical">

    <EditText
        android:id="@+id/emailTo"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter recipient email"
        android:inputType="textEmailAddress"/>

    <EditText
        android:id="@+id/emailSubject"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter subject"
        android:inputType="text"/>

    <EditText
        android:id="@+id/emailMessage"
```



```

        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter message"
        android:inputType="textMultiLine"
        android:minHeight="100dp"/>

        <Button
            android:id="@+id/sendEmail"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Send Email"
            android:layout_margin="10dp"/>
    </LinearLayout>

```

MainActivity.java File:

```

package com.asha.email;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText emailTo, emailSubject, emailMessage;
    Button sendEmail;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize UI components
        emailTo = findViewById(R.id.emailTo);
        emailSubject = findViewById(R.id.emailSubject);
        emailMessage = findViewById(R.id.emailMessage);
        sendEmail = findViewById(R.id.sendEmail);

        // Button click listener
        sendEmail.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                sendEmail();
            }
        });
    }

    // Method to send email
    private void sendEmail() {
        String recipient = emailTo.getText().toString().trim();
        String subject = emailSubject.getText().toString().trim();
        String message = emailMessage.getText().toString().trim();

        // Validate input
    }

```

```

        if (recipient.isEmpty()) {
            Toast.makeText(this, "Please enter recipient email",
Toast.LENGTH_SHORT).show();
            return;
        }
        if (!android.util.Patterns.EMAIL_ADDRESS.matcher(recipient).matches()) {
            Toast.makeText(this, "Enter a valid email address",
Toast.LENGTH_SHORT).show();
            return;
        }
        if (subject.isEmpty()) {
            Toast.makeText(this, "Please enter a subject", Toast.LENGTH_SHORT).show();
            return;
        }
        if (message.isEmpty()) {
            Toast.makeText(this, "Please enter a message", Toast.LENGTH_SHORT).show();
            return;
        }

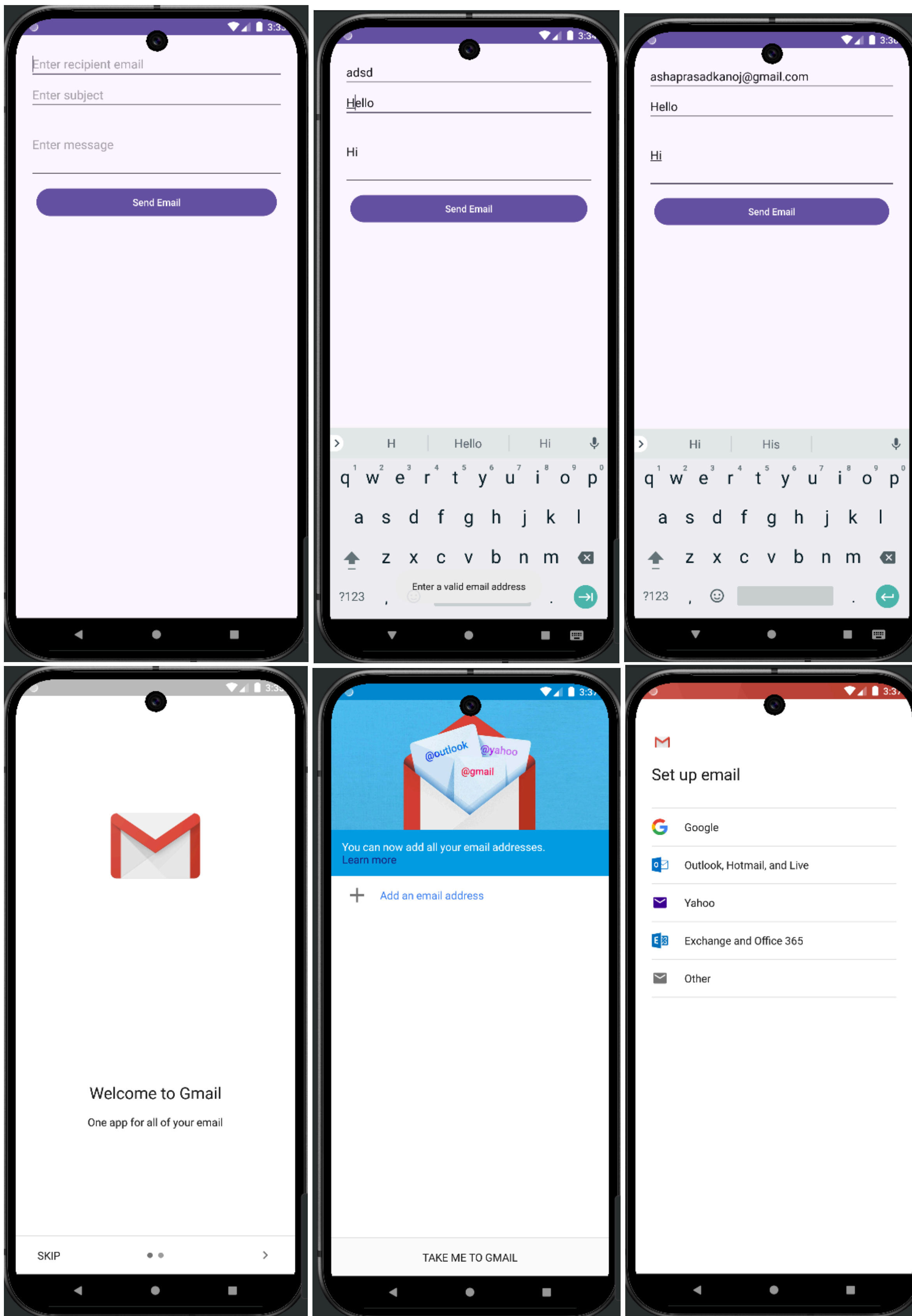
        // Open email app with pre-filled details
        Intent intent = new Intent(Intent.ACTION_SENDTO);
        intent.setData(Uri.parse("mailto:" + recipient));
        intent.putExtra(Intent.EXTRA_SUBJECT, subject);
        intent.putExtra(Intent.EXTRA_TEXT, message);

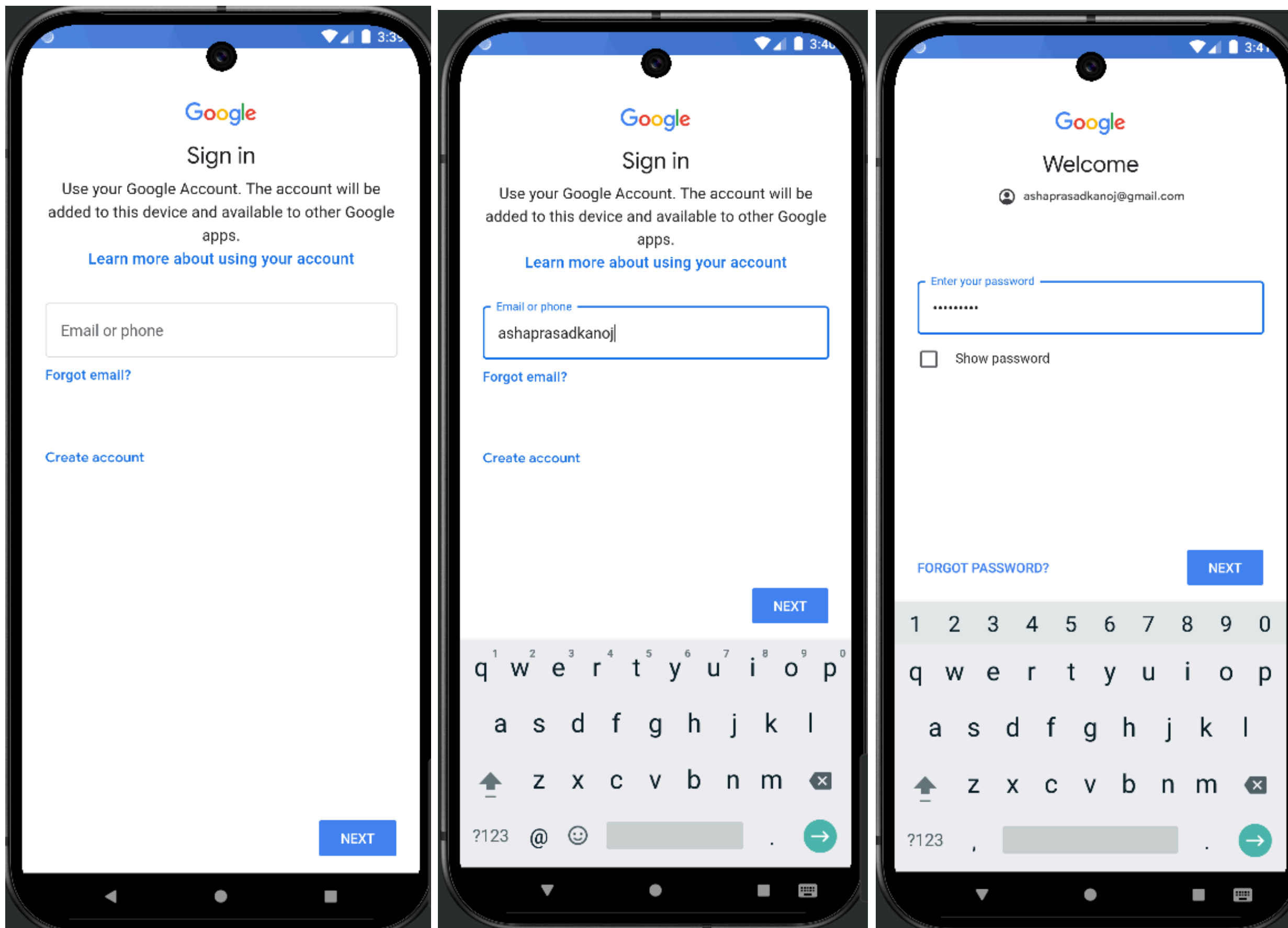
// if mail subject & message is not taken directly after providing input by user, add this
code
//intent.setType("message/rfc822");
// ensures only email apps handle this
//intent.putExtra(Intent.EXTRA_EMAIL, new String[]{recipient});

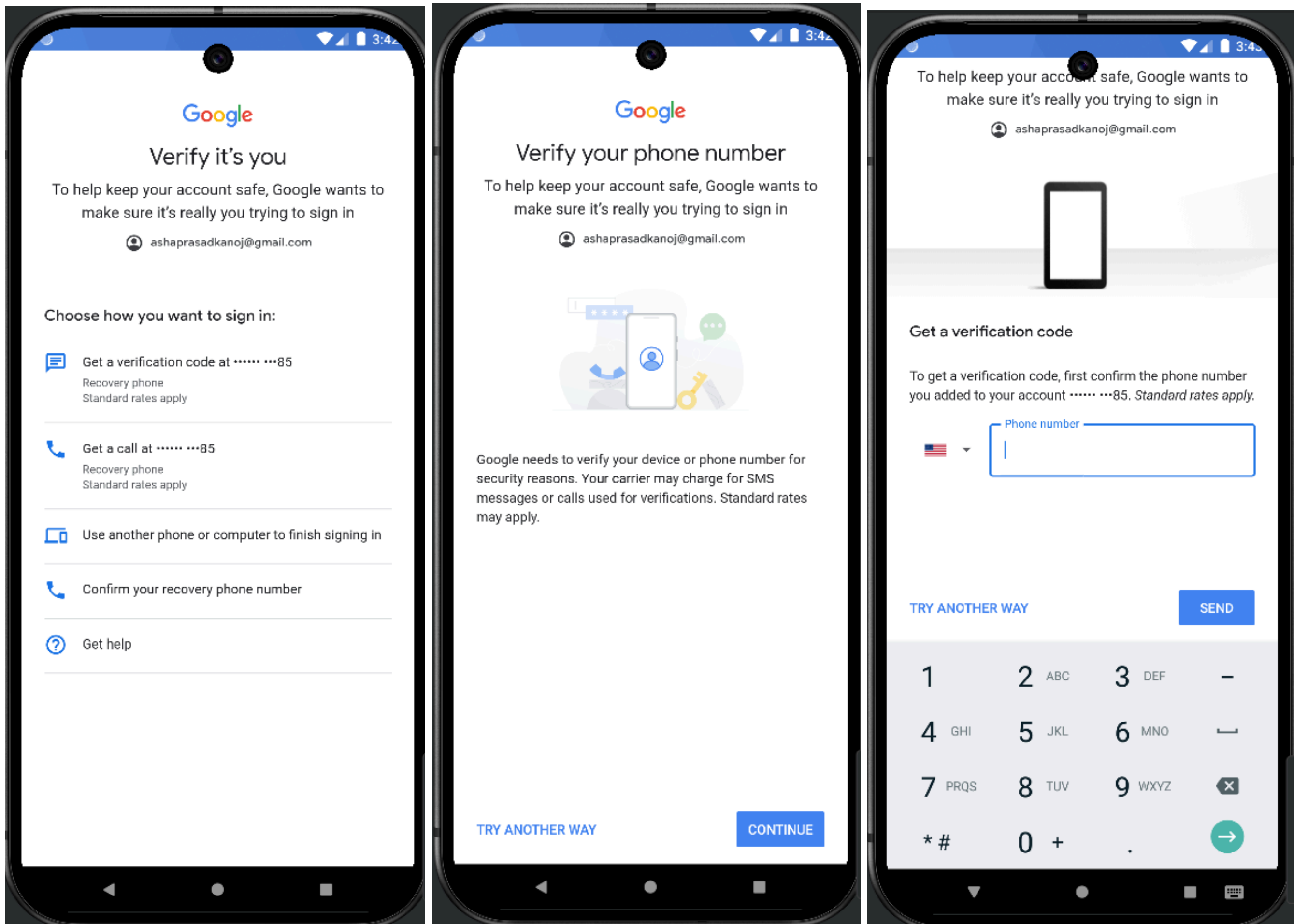
        try {
            startActivity(Intent.createChooser(intent, "Choose an Email Client"));
        } catch (Exception e) {
            Toast.makeText(this, "No email client installed", Toast.LENGTH_SHORT).show();
        }
    }
}

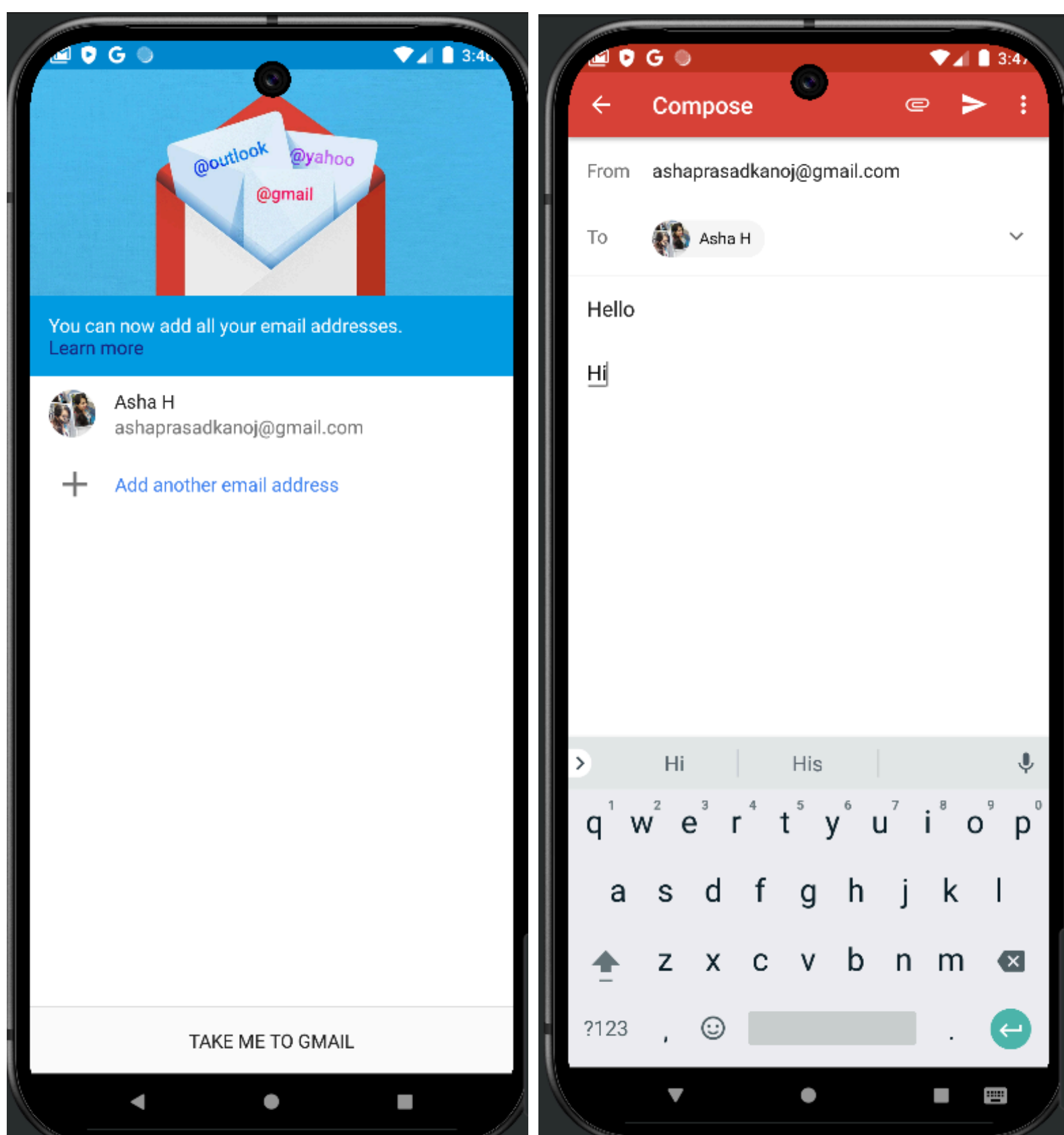
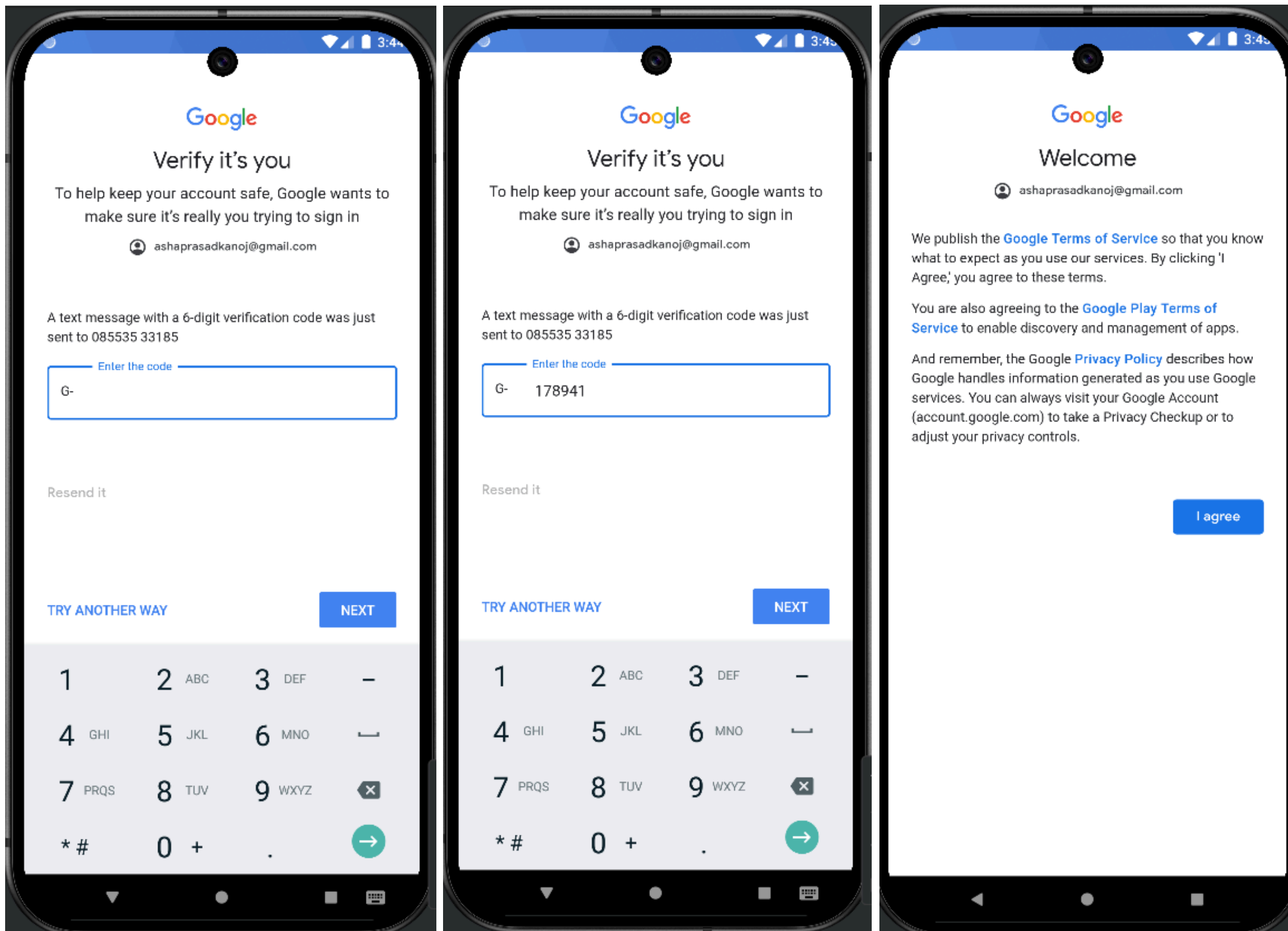
```

Output:





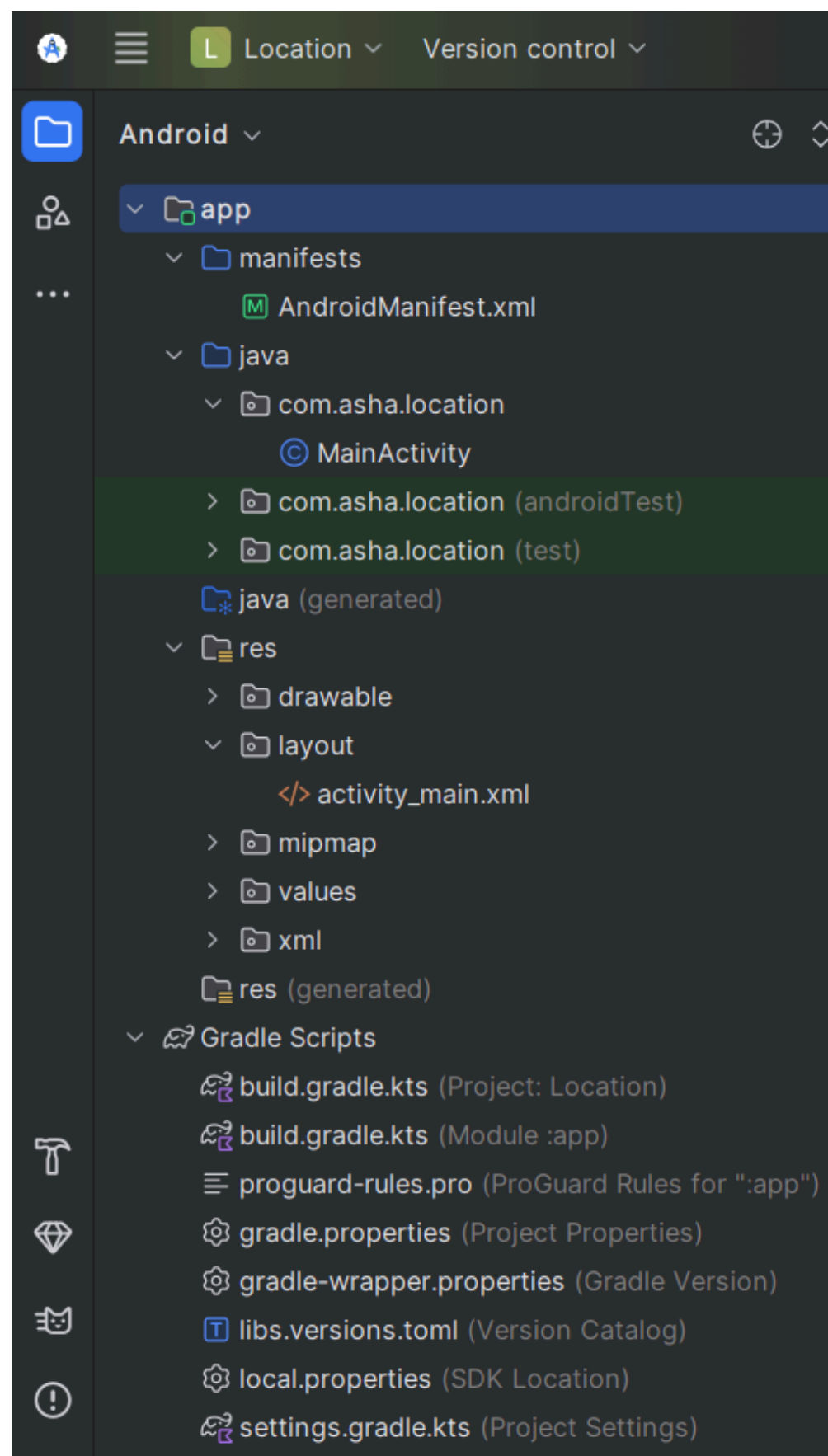




13. Display Map based on the Current/given location.

API KEY: AlzaSyCFSGQdjFJ9tilySMXDDbOd7Cho3JFWg5Q

Structure of a Program:



AndroidManifest.xml File:

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"/>
    <uses-permission android:name="android.permission.INTERNET"/>
    <uses-permission android:name="android.permission.ACCESS_NETWORK_STATE"/>
    <uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION"/>

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Location"
        tools:targetApi="31">
```

```

        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>

        <meta-data
            android:name="com.google.android.geo.API_KEY"
            android:value="AIzaSyCFSGQdjFJ9tilySMXDDb0d7Cho3JFWg5Q" />

    </application>
</manifest>

```

Activity_main.xml File:

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <fragment
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:id="@+id/id_map"
        android:name="com.google.android.gms.maps.SupportMapFragment"/>

</RelativeLayout>

```

Build.gradle.kts(module:app)

```

dependencies {

    implementation(libs.appcompat)
    implementation(libs.material)
    implementation(libs.activity)
    implementation(libs.constraintlayout)
    testImplementation(libs.junit)
    androidTestImplementation(libs.ext.junit)
    androidTestImplementation(libs.espresso.core)
    implementation("androidx.appcompat:appcompat:1.6.1")
    implementation("com.google.android.material:material:1.11.0")
    implementation("androidx.constraintlayout:constraintlayout:2.1.4")
    testImplementation("junit:junit:4.13.2")
    androidTestImplementation("androidx.test.ext:junit:1.1.5")
    androidTestImplementation("androidx.test.espresso:espresso-core:3.5.1")
    implementation("com.google.android.gms:play-services-maps:18.2.0")
    implementation("com.google.android.gms:play-services-location:21.2.0")
}

```

Note: Once you add dependencies sync the project.

MainActivity.java File:

```

package com.asha.location;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import com.google.android.gms.maps.CameraUpdateFactory;
import com.google.android.gms.maps.GoogleMap;
import com.google.android.gms.maps.OnMapReadyCallback;
import com.google.android.gms.maps.SupportMapFragment;
import com.google.android.gms.maps.model.LatLng;
import com.google.android.gms.maps.model.MarkerOptions;

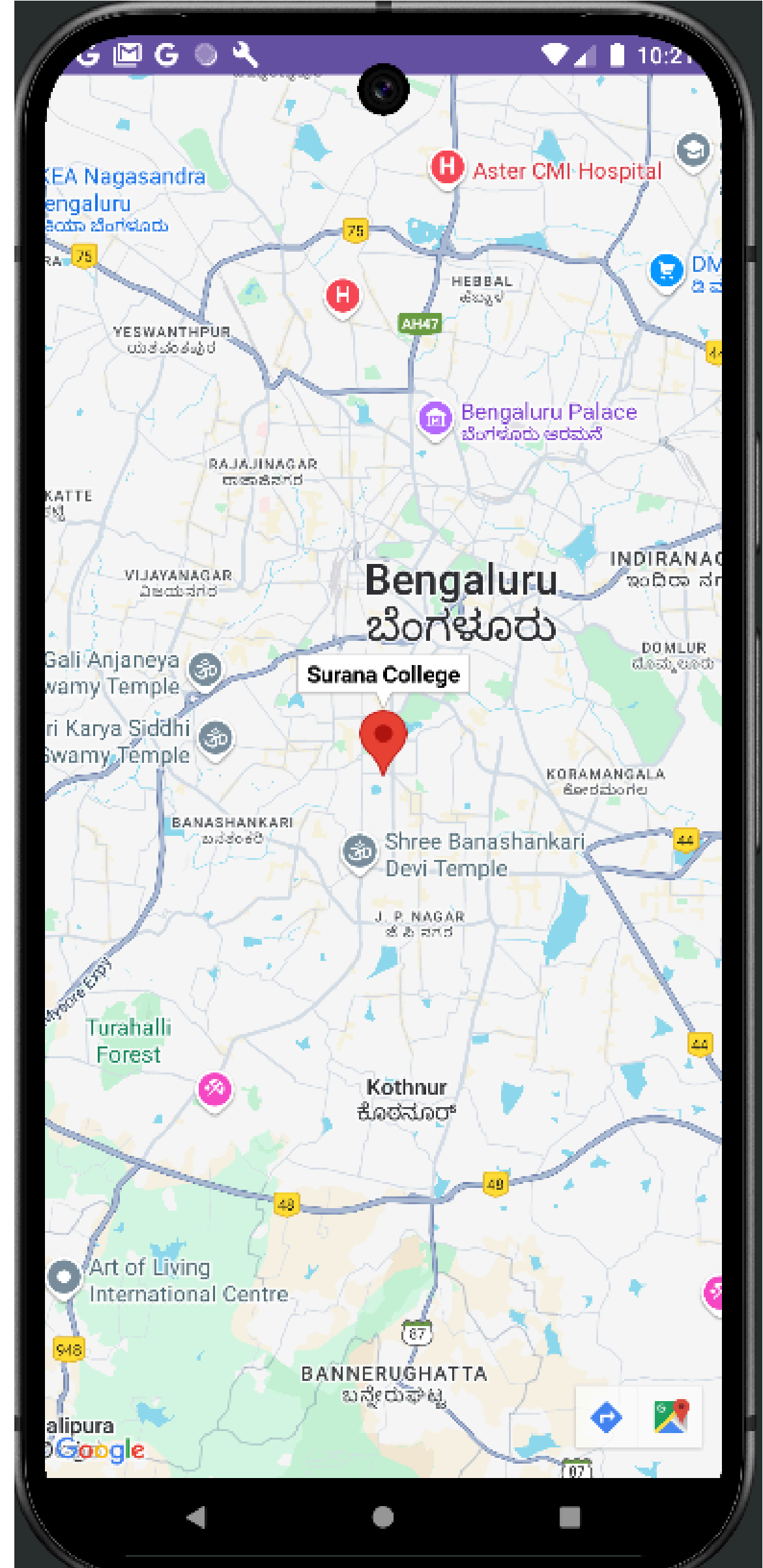
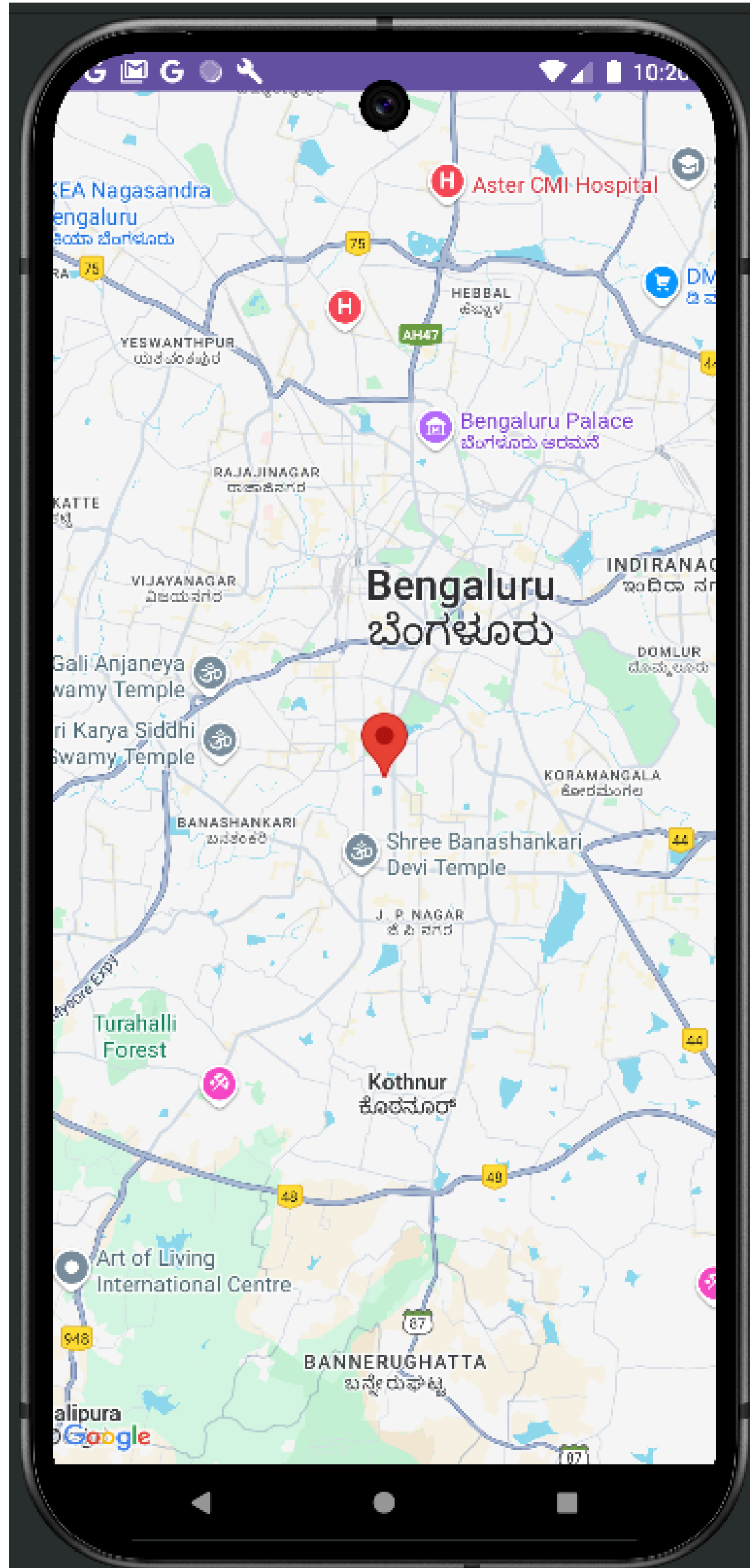
import android.os.Bundle;

public class MainActivity extends AppCompatActivity implements OnMapReadyCallback {
    private GoogleMap mMap;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        SupportMapFragment mapFragment = (SupportMapFragment)
            getSupportFragmentManager()
                .findFragmentById(R.id.id_map);
        mapFragment.getMapAsync(this);
    }
    @Override
    public void onMapReady(@NonNull GoogleMap googleMap) {
        LatLng location=new LatLng(12.9364 , 77.5782 );
        // co-ordinates of surana college 12.9364° N, 77.5782° E
        googleMap.addMarker(new MarkerOptions().position(location).title("Surana
College"));
        googleMap.moveCamera(CameraUpdateFactory.newLatLngZoom(location,12));
    }
}

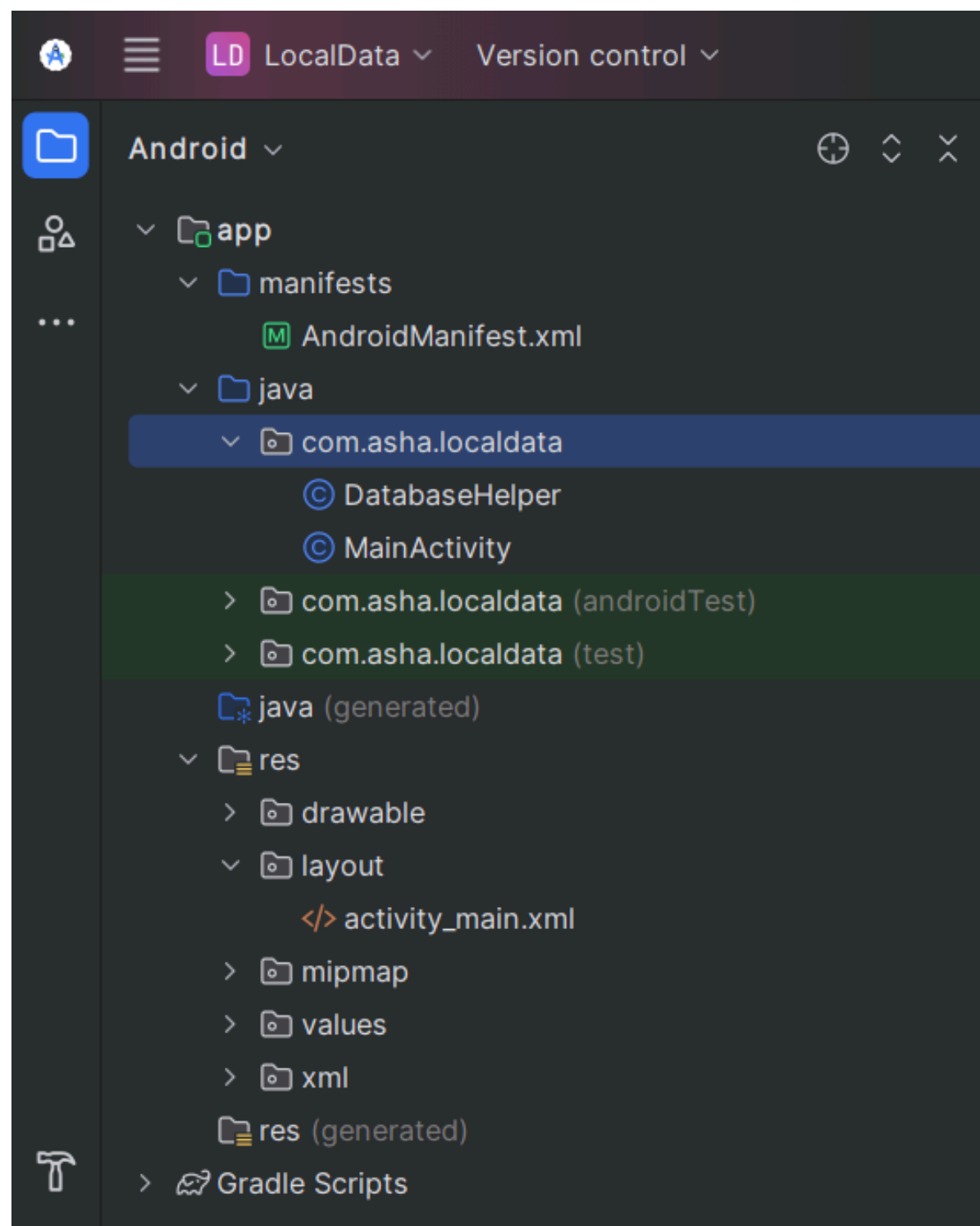
```


Output:



14. Create / Read / Write data with database (SQLite).

Structure of Program



AndroidManifest.xml File:

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
    <uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE"/>
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.LocalData"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>
```

```

        </activity>
    </application>

</manifest>

```

activity_main.xml File:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp"
    android:orientation="vertical">

    <EditText
        android:id="@+id/name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Name"/>

    <EditText
        android:id="@+id/age"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Age"
        android:inputType="number"/>

    <Button
        android:id="@+id/save"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Save Data"/>

    <Button
        android:id="@+id/retrieve"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Retrieve Data"/>

    <TextView
        android:id="@+id/result"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Stored Data Will Appear Here"
        android:padding="10dp"/>

</LinearLayout>

```

MainActivity.java File:

```

package com.asha.localdata;

import android.content.SharedPreferences;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

```



```

import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText Name, Age;
    Button Save, Retrieve;
    TextView Result;
    DatabaseHelper databaseHelper;
    SharedPreferences sharedPreferences;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initializing UI components
        Name = findViewById(R.id.name);
        Age = findViewById(R.id.age);
        Save = findViewById(R.id.save);
        Retrieve = findViewById(R.id.retrieve);
        Result = findViewById(R.id.result);

        // Initializing SQLite database and SharedPreferences
        databaseHelper = new DatabaseHelper(this);
        sharedPreferences = getSharedPreferences("MyPrefs", MODE_PRIVATE);

        // Save button click event
        Save.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String name = Name.getText().toString().trim();
                String ageStr = Age.getText().toString().trim();

                // Validate inputs
                if (name.isEmpty() || ageStr.isEmpty()) {
                    Toast.makeText(MainActivity.this, "Please enter both Name and Age",
Toast.LENGTH_SHORT).show();
                    return;
                }

                try {
                    int age = Integer.parseInt(ageStr);

                    // Save data to SQLite
                    boolean isInserted = databaseHelper.insertUser(name, age);

                    // Save data to SharedPreferences
                    SharedPreferences.Editor editor = sharedPreferences.edit();
                    editor.putString("name", name);
                    editor.putInt("age", age);
                    editor.apply();

                    if (isInserted) {
                        Toast.makeText(MainActivity.this, "Data Saved Successfully!",
Toast.LENGTH_SHORT).show();
                        Name.setText(""); // Clear input fields after saving
                        Age.setText("");
                    } else {

```



```

        Toast.makeText(MainActivity.this, "Failed to Save Data",
Toast.LENGTH_SHORT).show();
    }

    } catch (NumberFormatException e) {
        Toast.makeText(MainActivity.this, "Invalid Age Input",
Toast.LENGTH_SHORT).show();
    }
}

});

// Retrieve button click event
Retrieve.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        // Retrieve data from SharedPreferences
        String sharedName = sharedPreferences.getString("name", "No Name Found");
        int sharedAge = sharedPreferences.getInt("age", 0);

        // Retrieve data from SQLite
        String sqliteData = databaseHelper.getAllUsers();
        if (sqliteData.isEmpty()) {
            sqliteData = "No data found in SQLite Database.";
        }

        // Display retrieved data
        Result.setText("From SharedPreferences:\nName: " + sharedName + ", Age: "
+ sharedAge + "\n\n" + "From SQLite Database:\n" + sqliteData);
    }
});
}
}

```

DatabaseHelper.java File:

```

package com.asha.localdata;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import android.util.Log;

public class DatabaseHelper extends SQLiteOpenHelper {

    private static final String DATABASE_NAME = "UserData.db";
    private static final String TABLE_NAME = "users";
    private static final String COL_ID = "id";
    private static final String COL_NAME = "name";
    private static final String COL_AGE = "age";

    public DatabaseHelper(Context context) {
        super(context, DATABASE_NAME, null, 1);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {

```

```

String createTable = "CREATE TABLE " + TABLE_NAME + " (" +
    COL_ID + " INTEGER PRIMARY KEY AUTOINCREMENT, " +
    COL_NAME + " TEXT, " +
    COL_AGE + " INTEGER)";
db.execSQL(createTable);
}

@Override
public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
    db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
    onCreate(db);
}

public boolean insertUser(String name, int age) {
    SQLiteDatabase db = this.getWritableDatabase();
    ContentValues contentValues = new ContentValues();
    contentValues.put(COL_NAME, name);
    contentValues.put(COL_AGE, age);
    long result = db.insert(TABLE_NAME, null, contentValues);
    return result != -1;
}

public String getAllUsers() {
    SQLiteDatabase db = this.getReadableDatabase();
    Cursor cursor = db.rawQuery("SELECT * FROM " + TABLE_NAME, null);
    StringBuilder result = new StringBuilder();
    while (cursor.moveToNext()) {
        result.append("ID: ").append(cursor.getInt(0))
            .append(", Name: ").append(cursor.getString(1))
            .append(", Age: ").append(cursor.getInt(2))
            .append("\n");
    }
    cursor.close();
    return result.toString();
}
}

```

Output: